

AN EXPLICIT FORMULA FOR THE LODAY ASSEMBLY

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ABSTRACT. We give an explicit description of the Loday assembly map on homotopy groups when restricted to a certain subgroup coming from the Atiyah-Hirzebruch spectral sequence. This proves and generalises a formula about the Loday assembly map on the first homotopy group written down in a survey article by Lück and Reich. Furthermore, we use this new result to prove the Loday assembly for the integral group ring of cyclic group of order 2 is a bijection on the second homotopy group. Finally, we show that our formula can be used to compute the universal assembly map on homotopy.

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1. INTRODUCTION

Classically, algebraic K-theory deals with three Abelian groups $K_0(R)$, $K_1(R)$ and $K_2(R)$ associated to a ring R . See [Ros95] for their definitions. For a topological space X , topologists have found geometric applications of the groups $K_i(\mathbb{Z}[\pi_1(X)])$ for $i = 0, 1, 2$, where $\mathbb{Z}[\pi_1(X)]$ is the integral group ring of the fundamental group of X . For example, an important application of K_0 was found by Wall [Wal65]. He defined an element $\chi(X)$ in a quotient group $\text{Wh}_0(\pi_1(X))$ of $K_0(\mathbb{Z}[\pi_1(X)])$, known as *Wall's finiteness obstruction*, and showed that a sufficiently nice X is homotopy equivalent to a finite CW-complex if and only if $\chi(X) = 0$. (See [Ros05, Theorem 1 on page 579] for the precise statement.) The second example, which is due to Whitehead, is an attempt to classify manifolds of dimension at least five. In a collection of work [Whi39; Whi41; Whi50], he defined the *Whitehead torsion* $\tau(f) \in \text{Wh}_1(\pi_1(X))$ of a continuous map f , where $\text{Wh}_1(\pi_1(X))$ is a quotient of $K_1(\mathbb{Z}[\pi_1(X)])$. In particular, the vanishing of the Whitehead torsion allowed Smale to prove the Poincaré Conjecture in dimensions greater than four [Sma61].

For an ideal $I \subseteq R$, there is a natural exact sequence

$$K_2(R) \rightarrow K_2(R/I) \rightarrow K_1(R, I) \rightarrow K_1(R) \rightarrow K_1(R/I) \rightarrow K_0(R, I) \rightarrow K_0(R) \rightarrow K_0(R/I) \quad (1.0.1)$$

(see [Ros95, Theorem 4.3.1 on page 200]), and a lot of effort went into the search for definitions for $K_i(R)$ with $i \geq 3$ to extend this exact sequence to the left. The first widely accepted definition was due to Quillen [Qui72; Qui73], for which he was awarded the Fields Medal in 1978. His idea was that one should try to construct the groups $K_i(R)$ not one at a time but all at once, as homotopy groups

$$K_i(R) = \pi_i(K(R)),$$

of a certain topological space $K(R)$ constructed to reflect structures of the category of finitely generated projective R -modules. It was later discovered by Gersten and Wagoner that the space $K(R)$ is an infinite loop space [Ger72; Wag72], and hence is the 0-th space of an Ω -spectrum $\mathbb{K}_R^{\text{GW}}$ (see Definition 2.13), making algebraic K-theory part of stable homotopy theory. Several definitions of algebraic K-theory came out to account for this property, notably Waldhausen's $\mathbf{S}_\bullet$ -construction [Wal78c]. Nowadays, people understand algebraic K-theory as a machine that takes in nice categorical data to produce Ω -spectra, or more precisely, a functor

$$\mathbb{K} : \text{SymMonCat} \rightarrow \Omega\text{-Spectra} \quad (1.0.2)$$

from symmetric monoidal categories to Ω -spectra such that the homotopy groups

$$\pi_i \left(\mathbb{K} \left(\mathcal{P}\text{roj}_R^{\text{fg}} \right) \right)$$

of the Ω -spectrum, obtained by evaluating at the category $\mathcal{P}\text{roj}_R^{\text{fg}}$ of finitely generated projective R -modules, recover the classical K-groups of R for $i = 0, 1, 2$ and the higher K-groups defined by Quillen.

However, the general consensus is that computing homotopy groups is difficult. Hence, investigating algebraic K-theory of rings is not an easy task. On the other hand, homology is more accessible—there are excisions, the Mayer-Vietoris sequence, and even spectral sequences in homology that are more user-friendly than they would be in homotopy. This is the story of assembly—to approximate homotopy theory by a generalised homology theory.

1.1. History and Motivations. In his dissertation [Lod76], Loday defined a map

$$\alpha_{\text{Loday}} : BG_+ \wedge \mathbb{K}_R^{\text{GW}} \rightarrow \mathbb{K}_{R[G]}^{\text{GW}} \quad (1.1.1)$$

of spectra, which is now known as the Loday assembly (see Definition 3.3), to unify the classical Whitehead groups $\text{Wh}_i(G)$ for $i = 0, 1, 2$ studied by Wall [Wal65], Whitehead [Whi39; Whi41; Whi50], and Hatcher-Wagoner [HW73]. These groups are isomorphic to the cokernel

$$\text{Wh}_i(G) \cong \text{coker} \left(\pi_i(\alpha_{\text{Loday}}) : \pi_i(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \rightarrow K_i(\mathbb{Z}[G]) \right) \quad (1.1.2)$$

for $i = 0, 1, 2$. See [Lod76, page 357–364], or Corollary 3.10 for the case $i = 1$. Before this work appeared, Wall defined a version of assembly

$$A_i : H_i(BG; \mathbb{Q}) \rightarrow L_i(\mathbb{Z}[G]) \otimes \mathbb{Q} \quad (1.1.3)$$

for L-theory. Furthermore, he showed that the injectivity of A_i implies the classical Novikov Conjecture about homotopy invariance of the higher signature of a closed, oriented manifold M when $G = \pi_1(M)$ [Wal70, 17H]. The converse is also true [KM81], and therefore the classical Novikov Conjecture is equivalent to the injectivity of the map A_i for all i . Motivated by this, Hsiang cast the following conjecture in his 1984 ICM address:

Conjecture 1.1 (K-theoretic Novikov Conjecture, [Hsi84])

Let R be a regular ring, and G be a torsion-free group. Then the map

$$\pi_i(\alpha_{\text{Loday}}) \otimes \text{id}_{\mathbb{Q}} : \pi_i(BG_+ \wedge \mathbb{K}_R^{\text{GW}}) \otimes \mathbb{Q} \rightarrow K_i(R[G]) \otimes \mathbb{Q} \quad (1.1.4)$$

is injective for all i .

Attempts on answering the K-theoretic Novikov Conjecture 1.1 often leads to the creation of new mathematics, most notably the Topological Cyclic Homology, which allowed Bökstedt-Hsiang-Madsen to prove the following:

Theorem 1.2 ([BHM93, Theorem 9.13 on page 535])

The K-theoretic Novikov Conjecture 1.1 is true if the ordinary homology groups of the classifying space BG for G are all finitely generated.

A stronger statement is also possible:

Conjecture 1.3 (Classical Farrell-Jones Conjecture)

Let R be a regular ring, and G be a torsion-free group. Then the map

$$\pi_i(\alpha_{\text{Loday}}) : \pi_i(BG_+ \wedge \mathbb{K}_R^{\text{GW}}) \rightarrow K_i(R[G]) \quad (1.1.5)$$

is an isomorphism for all i .

This conjecture can be understood as a conceptual approach to computing algebraic K-theory of a group ring via a homology theory, but it has important implications for a range of topics, notably the Borel Conjecture concerning the topological rigidity of closed spherical manifolds, and the Kaplansky Conjecture about the idempotents in a group ring.

The following case has been verified:

Theorem 1.4 ([BLR08, Theorem 1.1 (i) on page 58])

The [Classical Farrell-Jones Conjecture 1.3](#) is true if the group G is word-hyperbolic.

The work presented here is motivated by the [Classical Farrell-Jones Conjecture 1.3](#).

1.2. What Are We Trying to Do? The purpose of this dissertation is to help the author to obtain his doctoral degree by studying the Loday assembly.

More precisely, the source of the Loday assembly

$$\alpha_{\text{Loday}} : BG_+ \wedge \mathbb{K}_R^{\text{GW}} \rightarrow \mathbb{K}_{R[G]}^{\text{GW}}$$

represents a generalised homology theory. Hence, its homotopy groups can be computed by the Atiyah-Hirzebruch spectral sequence

$$\begin{aligned} E_{p,q}^2 &\cong H_p(BG_+; K_q(R)) \\ &\Rightarrow \pi_{p+q}(BG_+ \wedge \mathbb{K}_R^{\text{GW}}). \end{aligned}$$

From the construction of the spectral sequence, we know there is a subgroup $E_{1,i}^\infty$ of the homotopy group $\pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\text{GW}})$ coming from the 1-skeleton of the classifying space BG . We provide a formula for the restriction $\pi_{i+1}(\alpha_{\text{Loday}})|_{E_{1,i}^\infty}$:

Theorem 1.5 (See [Theorem 3.8](#))

Let R be a ring and G be a group. The filler of the diagram

$$\begin{array}{ccccc} E_{1,i}^2 & \cong & H_1(BG; K_i(R)) & \cong & G_{ab} \otimes K_i(R) \xrightarrow{\Phi_i} K_{i+1}(R[G]) \\ \downarrow & & & & \uparrow \pi_{i+1}(\alpha_{\text{Loday}}) \\ E_{1,i}^\infty & \hookrightarrow & \xrightarrow{\hspace{10em}} & \pi_{i+1}(BG \wedge \mathbb{K}_R^{\text{GW}}) & \end{array} \quad (1.2.1)$$

is induced by the group homomorphism

$$\begin{aligned} G \times K_i(R) &\rightarrow K_{i+1}(R[G]) \\ (g, [f]) &\mapsto [f] \star \{g^{-1}\}, \end{aligned}$$

for which on the left-hand side, the element g^{-1} is considered as an element in $GL(1, R[G])$, and $\star$ is the Loday product.

The derivation of our formula involves extending the original Loday pairing

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \rightarrow BGL(R \otimes S)^+ \quad (1.2.2)$$

to the full K-theory space

$$\gamma'_{\text{Loday}} : [K_0(R) \times BGL(R)^+] \wedge [K_0(S) \times BGL(S)^+] \rightarrow K_0(R \otimes S) \times BGL(R \otimes S)^+ \quad (1.2.3)$$

in a non-obvious way (see [Definition 2.5](#)). Here, the superscript “+” denotes Quillen’s plus construction of the classifying space $BGL(R)$ relative to the subgroup $E(R)$ generated by elementary matrices. In particular,

- (i) we relate the product map

$$\star' : K_i(R) \otimes K_j(S) \rightarrow K_{i+j}(R \otimes S)$$

induced by the extend pairing γ'_{Loday} to the classical product maps defined by Milnor in [\[Mil72\]](#) (see [Theorem 2.9](#)).

- (ii) We recover (and generalise) a formula written down in a survey article [\[LR05\]](#) by Lück-Reich about the Loday assembly on π_1 (see [Corollary 3.9](#)).
- (iii) We obtain injectivity/bijectivity results on π_2 for the group ring $\mathbb{Z}[C_2^{\oplus n}]$ (see [Example 4.5](#) and [Example 4.10](#)).
- (iv) We prove that the non-injectivity phenomenon of the Loday assembly on π_2 is related to deciding whether the Steinberg symbol $\{-1, g\}_{\text{St}} \in \text{St}(\mathbb{Z}[G])$ is trivial or not (see [Theorem 4.6](#)).
- (v) We show that our formula can be used to compute the universal assembly in the sense of Weiss-Williams

$$\hat{\alpha}_{BG} : BG_+ \wedge \mathbb{K}_R^{\text{PW}} \rightarrow \mathbb{K}_{R[G]}^{\text{PW}}$$

written in terms of the non-connective Pedersen-Weibel K-theory spectrum $\mathbb{K}_R^{\text{PW}}$ of a ring R (see [Theorem 5.49](#)).

We elaborate more on [item \(ii\)](#). The assembly on fundamental groups

$$\pi_1(\hat{\alpha}_{BG}) : \pi_1(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{PW}}) \rightarrow K_1(\mathbb{Z}[G]) \quad (1.2.4)$$

was first described by Waldhausen. It is induced by the group homomorphism

$$\begin{aligned} \{\pm 1\} \times G &\rightarrow K_1(\mathbb{Z}[G]) \\ (\pm 1, g) &\mapsto \{\pm g\} \end{aligned} \quad (1.2.5)$$

under the identification

$$\begin{aligned} \pi_1(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) &\cong K_1(\mathbb{Z}) \oplus H_1(G; K_0(\mathbb{Z})) \\ &\cong K_1(\mathbb{Z}) \oplus [G_{ab} \otimes K_0(\mathbb{Z})] \\ &\cong \{\pm 1\} \oplus G_{ab} \end{aligned} \quad (1.2.6)$$

resulting from the vanishing of differentials in the Atiyah-Hirzebruch spectral sequence [\[Wal78b, Assertion 15.8 on page 229\]](#). On the left-hand side of [Equation \(1.2.5\)](#), we think

of $\{\pm g\}$ is represented by $\pm g \in GL(1, \mathbb{Z}[G])$. A formula for arbitrary regular ring was later written down in the survey article by Lück-Reich [LR05, page 708].

To prove (1.2.4) is as given by (1.2.5), Waldhausen reformulates the Loday assembly in terms of Quillen's Q-construction, and interprets the source of the assembly as a generalised homology theory for simplicial sets. Then, he verifies the formula on the simplicial level. The work presented here provides a different approach to prove Waldhausen's formula by working in Loday's original setting—using Quillen's plus construction. This new approach allows us to get higher degree results, and in particular, an elegant formula on π_2 (see Corollary 4.4).

1.3. Organisation. This dissertation consists of five parts, including the introduction. In Chapter 2, we review the construction of the original Loday pairing map and extend it to the full K-theory space using the Gersten-Wagoner delooping. We then relate the induced product maps on the K-groups with the classical product maps defined by Milnor and use the extended Loday pairing to construct the non-connective Gersten-Wagoner Algebraic K-theory spectrum $\mathbb{K}_R^{\text{GW}}$ of a ring R .

In Chapter 3, we use the extended Loday pairing to define the Loday assembly and identify the subgroup of the source that we are interested in by studying the Atiyah-Hirzebruch spectral sequence. It also contains the statement and the proof of our main result.

Chapter 4 deals with the injectivity problem for the Loday assembly on π_2 . We study the extension problem for the second homotopy group of the source of the Loday assembly coming from the Atiyah-Hirzebruch spectral sequence. Then we provide injectivity/bijection results for the Loday assembly for n -fold direct sum of the cyclic group of order 2. We also show that non-injectivity phenomenon of the Loday assembly on π_2 is related to deciding whether the Steinberg symbol $\{-1, g\}_{\text{St}} \in \text{St}(\mathbb{Z}[G])$ is trivial or not.

Chapter 5 is a quick summary of Sperber's work on proving the Loday assembly is the universal assembly [Spe04]. We show that our version of the Loday assembly can be used to compute the universal assembly on homotopy even with the non-obvious extension the original Loday pairing. We begin by discussing a model for the universal assembly

$$\widehat{\alpha}_{BG} : BG_+ \wedge \mathbb{K}_R^{\text{PW}} \rightarrow \mathbb{K}_{R[G]}^{\text{PW}} \quad (1.3.1)$$

written in terms of the Pedersen-Weibel K-theory spectrum (see Definition 5.42). Then, we construct two intermediate spectra and their versions of assembly

$$\alpha_{\text{proj}} : BG_+ \wedge \mathbb{K}_R^{\text{proj}} \rightarrow \mathbb{K}_{R[G]}^{\text{proj}} \quad (1.3.2)$$

$$\alpha_{\text{free}} : BG_+ \wedge \mathbb{K}_R^{\text{free}} \rightarrow \mathbb{K}_{R[G]}^{\text{free}} \quad (1.3.3)$$

(see Definition 5.44), which allows us to move from a theoretically-friendly model for the assembly to a computation-friendly model. More specifically, there is zig-zag diagram

$$\begin{array}{ccc}
BG_+ \wedge \mathbb{K}_R^{\text{PW}} & \xrightarrow{\widehat{\alpha}_{BG}} & \mathbb{K}_{R[G]}^{\text{PW}} \\
\text{id} \wedge (5.5.9) \downarrow & & \downarrow (5.5.9) \\
BG_+ \wedge \mathbb{K}_R^{\text{proj}} & \xrightarrow{\alpha_{\text{proj}}} & \mathbb{K}_{R[G]}^{\text{proj}} \\
\text{id} \wedge (5.5.6) \uparrow & & \uparrow (5.5.6) \\
BG_+ \wedge \mathbb{K}_R^{\text{free}} & \xrightarrow{\alpha_{\text{free}}} & \mathbb{K}_{R[G]}^{\text{free}}
\end{array} \tag{1.3.4}$$

of spectra that commutes up after passing to homotopy groups. We show that our formula for the restriction $\pi_{i+1}(\alpha_{\text{Loday}})|_{E_{1,i}^\infty}$ can be used to compute the restriction $\pi_{i+1}(\alpha_{\text{free}})|_{E_{1,i}^\infty}$ for $i \geq 2$ (see [Theorem 5.49](#)) and therefore, computes the universal assembly on higher homotopy when restricted onto the subgroup coming from the 1-skeleton of BG .

1.4. Notations and Conventions.

1.4.1. Rings and Modules. All *rings* in this article are associative, unital, but not necessarily commutative. A *module* over a ring will always be understood as left module.

Given a ring R , we define its *cone* to be the ring $\text{cone}(R)$ of locally finite matrices over R , that is,

$$\text{cone}(R) := \left\{ A \in M(R) \mid \begin{array}{l} \text{Each column and each row of } A \\ \text{has only finitely many non-zero entries} \end{array} \right\}. \tag{1.4.1}$$

There is an ideal $m(R)$ of $\text{cone}(R)$ consisting of infinite matrices with only finitely many non-zero entries. The *suspension* of R is defined to be the quotient

$$\Sigma R := \frac{\text{cone}(R)}{m(R)}. \tag{1.4.2}$$

We also define

$$\Sigma^0 R := R \tag{1.4.3}$$

and inductively

$$\Sigma^n R := \Sigma(\Sigma^{n-1} R). \tag{1.4.4}$$

Note that for any ring R , we have a natural isomorphism

$$\Sigma R \cong \Sigma \mathbb{Z} \otimes R.$$

Therefore, for any two rings R, S , we have

$$\Sigma R \otimes \Sigma S \cong \Sigma^2(R \otimes S). \tag{1.4.5}$$

1.4.2. *Topological Spaces and Spectra.* The term *topological space* (or simply *space*) will always mean a pointed topological space that is homotopy equivalent to a (not necessary finite) CW-complex. Maps between spaces are always pointed and continuous.

We say two maps $f, g : X \rightarrow Y$ of spaces are *weakly homotopic* if the restrictions $f|_K, g|_K$ are homotopic for every compact subset $K \subseteq X$. Weakly homotopic maps induce the same group homomorphisms on homotopy.

By *spectrum*, we mean a sequence $\mathbb{E} = \{\mathbb{E}_i \mid i \in I\}$ of spaces, where $I = \mathbb{N} \cup \{0\}$ or $\mathbb{Z}$, together with maps

$$f_i : \mathbb{S}^1 \wedge \mathbb{E}_i \rightarrow \mathbb{E}_{i+1}$$

for each $i \in I$, called the *structure maps*. An Ω -*spectrum* is a spectrum for which the adjoints

$$\bar{f}_i : \mathbb{E}_i \rightarrow \Omega \mathbb{E}_{i+1}$$

of the structure maps are all homotopy equivalences. A *morphism* or a *map* (resp. *weak morphism* or a *weak map*) $\varphi : \mathbb{E} \rightarrow \mathbb{F}$ of spectra is a collection $\{\varphi_i : \mathbb{E}_i \rightarrow \mathbb{F}_i \mid i \in I\}$ of maps that commutes with the structure maps (resp. up to weak homotopy).

If $\mathbb{E}, \mathbb{F}, \mathbb{G}$ are spectra, then a pairing (resp. weak pairing) $\mu : \mathbb{E} \wedge \mathbb{F} \rightarrow \mathbb{G}$ of spectra is a collection of maps $\mu_{m,n} : \mathbb{E}_m \wedge \mathbb{F}_n \rightarrow \mathbb{G}_{m+n}$ that commute with the structure maps up to homotopy (resp. weak homotopy). We abuse notation here to use the smash product $\mathbb{E} \wedge \mathbb{F}$ in writing down a pairing of spectra.

1.4.3. *Algebraic K-theory.* Let R be a ring. We write $K_0(R)$ to be *projective class group* of R , which is an Abelian group whose generators are isomorphism classes $[P]$ of finitely generated projective R -modules P , and whose relations are

$$[P_0] + [P_2] = [P_1] \tag{1.4.6}$$

for every short exact sequence

$$0 \rightarrow P_0 \rightarrow P_1 \rightarrow P_2 \rightarrow 0. \tag{1.4.7}$$

We write $BGL(R)^+$ to be *Quillen's plus construction* for the classifying space $BGL(R)$ for the general linear group $GL(R)$, related to the subgroup $E(R)$ generated by elementary matrices. For finite matrices, our convention is that $BGL(p, R)^+$ denotes $BGL(p, R)$ for $p \leq 2$; for $p \geq 3$, it denotes the plus construction of $BGL(p, R)$ relative to the subgroup $E(p, R)$ generated by elementary matrices.

The *K-theory space* of R is

$$K_R := K_0(R) \times BGL(R)^+, \tag{1.4.8}$$

where we think of $K_0(R)$ as a discrete group. In particular, the its homotopy groups $\pi_i(K_R)$ agrees with the classical K-groups for $i = 0, 1, 2$, and the higher K-groups defined by Quillen.

If $u \in GL(n, R)$, we write $\{u\} \in K_1(R)$ to be the class represented by u . If G is a group and $g \in G$, then we write $\{g\} \in K_1(R[G])$ to be the class represented by g , thinking it as an element in $GL(1, R[G])$.

2. LODAY PAIRING

2.1. The Loday Pairing. Let R, S be rings, and fix isomorphisms

$$\xi_{m,n} : R^m \otimes S^n \rightarrow (R \otimes S)^{mn}. \quad (2.1.1)$$

for every $m, n \in \mathbb{N} \cup \{0\}$. In this setting, the tensor product of matrices gives a group homomorphism

$$GL(m, R) \times GL(n, S) \rightarrow GL(mn, R \otimes S),$$

sending elementary matrices to elementary matrices. Hence we have an induced map

$$f_{m,n}^{R,S} : BGL(m, R)^+ \times BGL(n, S)^+ \rightarrow BGL(mn, R \otimes S)^+. \quad (2.1.2)$$

The convention is that $BGL(p, R)^+$ denotes $BGL(p, R)$ for $p \leq 2$; for $p \geq 3$, it denotes the plus construction of $BGL(p, R)$ relative to the subgroup $E(p, R)$ generated by elementary matrices. Write

$$i_{mn} : BGL(mn, R \otimes S)^+ \rightarrow BGL(R \otimes S)^+$$

to be map induced by the canonical inclusion

$$GL(mn, R \otimes S) \rightarrow GL(R \otimes S).$$

Definition 2.1 (The map $\gamma_{m,n}$, [Lod76, page 332])

The map

$$\gamma_{m,n} : BGL(m, R)^+ \times BGL(n, S)^+ \rightarrow BGL(R \otimes S)^+ \quad (2.1.3)$$

is defined by the formula:

$$\gamma_{m,n}(x, y) := i_{mn} \circ f_{m,n}^{R,S}(x, y) - i_{mn} \circ f_{m,n}^{R,S}(x, y_0) - i_{mn} \circ f_{m,n}^{R,S}(x_0, y). \quad (2.1.4)$$

Here, the minus sign on the right-hand side comes from the H-group structure of $BGL(R \otimes S)^+$; and x_0 (resp. y_0) is the base-point in $BGL(m, R)^+$ (resp. $BGL(n, S)^+$). (I.e., represented by the identity matrices.)

One gets a map

$$\gamma : BGL(R)^+ \times BGL(S)^+ \rightarrow BGL(R \otimes S)^+ \quad (2.1.5)$$

by letting $m, n \rightarrow \infty$. If we choose a different collection

$$\{\xi'_{m,n} : R^m \otimes S^n \rightarrow (R \otimes S)^{mn} \mid m, n \in \mathbb{N} \cup \{0\}\}$$

of isomorphisms, we get a different collection

$$\{\gamma'_{m,n} : BGL(m, R)^+ \times BGL(n, S)^+ \rightarrow BGL(R \otimes S)^+ \mid m, n \in \mathbb{N} \cup \{0\}\}$$

of continuous maps, and therefore, a different map

$$\gamma' : BGL(R)^+ \times BGL(S)^+ \rightarrow BGL(R \otimes S)^+$$

by letting $m, n \rightarrow \infty$. It turns out that the maps γ and γ' are weakly homotopic, that is, the restrictions $\gamma|_K$ and $\gamma'|_K$ are homotopic for every compact subset $K \subseteq BGL(R)^+ \times$

$BGL(S)^+$ (see [Lod76, Lemma 2.1.6 on page 333]). As a result, the choice of the collection $\{\xi_{m,n} \mid m, n \in \mathbb{N} \cup \{0\}\}$ becomes irrelevant after passing to homotopy groups.

The map γ is homotopy trivial on the wedge product $BGL(R)^+ \vee BGL(S)^+$. Hence, we obtain the following definition.

Definition 2.2 (The Loday pairing map γ_{Loday} , [Lod76, Section 2.1.7 on page 333])

The map

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \rightarrow BGL(R \otimes S)^+ \quad (2.1.6)$$

is defined to be the filler of the following homotopy commutative diagram:

$$\begin{array}{ccc} BGL(R)^+ \times BGL(S)^+ & \xrightarrow{\gamma} & BGL(R \otimes S)^+ \\ \text{proj} \downarrow & \nearrow \gamma_{\text{Loday}} & \\ BGL(R)^+ \wedge BGL(S)^+ & & \end{array} \quad (2.1.7)$$

This allows us to define a multiplicative structure on algebraic K-theory.

Definition 2.3 (Loday product $\star$, [Lod76, Section 2.1.10 on page 335])

For each integers $i, j \geq 1$, we define the product

$$\star : K_i(R) \otimes K_j(S) \rightarrow K_{i+j}(R \otimes S) \quad (2.1.8)$$

by

$$[f] \star [g] := [\gamma_{\text{Loday}} \circ (f \wedge g)], \quad (2.1.9)$$

where f and g are spheroids

$$\begin{aligned} f : \mathbb{S}^i &\rightarrow BGL(R)^+, \\ g : \mathbb{S}^j &\rightarrow BGL(S)^+. \end{aligned}$$

Proposition 2.4 (Properties of the Loday Product $\star$)

The product map $\star$ is

- (i) natural in R and S , associative and bilinear [Lod76, Theorem 2.1.11 on page 335];
- (ii) graded commutative: for every $[f] \in K_i(R)$ and $[g] \in K_j(S)$,

$$[f] \star [g] = (-1)^{ij} ([g] \star [f]) \in K_{i+j}(R \otimes S) \quad (2.1.10)$$

[Lod76, Theorem 2.1.12 on page 335];

- (iii) when $i = j = 1$, the product $\{u\} \star \{v\}$ is the inverse of the Steinberg symbol:

$$\{u\} \star \{v\} = -\{u, v\}_{\text{st}} \in K_2(R \otimes S) \quad (2.1.11)$$

[Lod76, Proposition 2.2.3 on page 337]. Here, we write the Abelian group $K_2(R \otimes S)$ additively.

2.2. The Extended Loday Pairing. We want to extend the Loday pairing map

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \rightarrow BGL(R \otimes S)^+$$

to a map

$$\gamma'_{\text{Loday}} : K_R \wedge K_S \rightarrow K_{R \otimes S},$$

where

$$K_R := K_0(R) \times BGL(R)^+. \quad (2.2.1)$$

This will yield an extended external product

$$K_i(R) \otimes K_j(S) \rightarrow K_{i+j}(R \otimes S)$$

for all $i, j \geq 0$. Moreover, this external product should be coherent with the isomorphism

$$K_i(R) \xrightarrow{\cong} K_{i+1}(\Sigma R) \quad (2.2.2)$$

promised by the **Gersten-Wagoner delooping** [Ger72; Wag72]:

$$K_R \simeq \Omega K_{\Sigma R}. \quad (2.2.3)$$

At this point, we need the closed form of the isomorphism in Equation (2.2.2). Define the element $\tau \in GL(\Sigma \mathbb{Z}) = GL(1, \Sigma \mathbb{Z})$ by

$$\tau := \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \cdots \\ 1 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 1 & 0 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \quad (2.2.4)$$

The isomorphism in Equation (2.2.2) is given by

$$\begin{cases} [f] \in K_i(R) \mapsto [f] \star \{\tau\} & \text{if } i \geq 1 \text{ ([Lod76, Corollary 2.3.6])}, \\ [P] \in K_0(R) \mapsto [P] \# \{\tau\} := \{p \otimes \tau + (1 - p) \otimes 1\} & \text{if } i = 0 \text{ ([Lod76, page 328])}, \end{cases} \quad (2.2.5)$$

where p is the projection operator associated to the finitely generated projective R -module P . Equally important, the extended external product should be related to the classical products

$$\# : K_i(R) \otimes K_j(S) \rightarrow K_{i+j}(R \otimes S) \quad (2.2.6)$$

defined by Milnor [Mil72] for $i, j \geq 0$, and $i + j \leq 2$.

Definition 2.5 (The Extended Loday pairing γ'_{Loday})
The map

$$\gamma'_{\text{Loday}} : K_R \wedge K_S \rightarrow K_{R \otimes S}$$

is defined to be the filler of the following homotopy commutative diagram:

$$\begin{array}{ccc} K_R \wedge K_S & \xrightarrow{\gamma'_{\text{Loday}}} & K_{R \otimes S} \\ \simeq \downarrow & & \downarrow \simeq \\ \Omega BGL(\Sigma R)^+ \wedge \Omega BGL(\Sigma S)^+ & \xrightarrow{\Omega^2 \gamma_{\text{Loday}}} & \Omega^2 BGL(\Sigma^2(R \otimes S))^+. \end{array} \quad (2.2.7)$$

The bottom arrow is given by sending the pair

$$(f : \mathbb{S}^1 \rightarrow BGL(\Sigma R)^+) \wedge (g : \mathbb{S}^1 \rightarrow BGL(\Sigma S)^+)$$

to the composition

$$\mathbb{S}^2 \cong \mathbb{S}^1 \wedge \mathbb{S}^1 \xrightarrow{f \wedge g} BGL(\Sigma R)^+ \wedge BGL(\Sigma S)^+ \xrightarrow{\gamma_{\text{Loday}}} BGL(\Sigma^2(R \otimes S))^+.$$

Furthermore, we define a new product map

$$\star' : K_i(R) \otimes K_j(S) \rightarrow K_{i+j}(R \otimes S) \quad (2.2.8)$$

for all $i, j \geq 0$ by

$$[f] \star' [g] := [\gamma'_{\text{Loday}} \circ (f \wedge g)]. \quad (2.2.9)$$

Corollary 2.6

Let R and S be rings. The pairing map

$$\gamma'_{\text{Loday}} : K_R \wedge K_S \rightarrow K_{R \otimes S}$$

defined in Definition 2.5 is

- (i) natural in R and S ;
- (ii) bilinear with respect to the H-group structures of the plus-constructions;
- (iii) associative;
- (iv) commutative;

up to weak homotopy.

The proof is exactly the same as [Lod76, Proposition 2.1.8 on page 334]. The two products are related in the following way:

Proposition 2.7

For each $i, j \geq 1$, if $[f] \in K_i(R)$ and $[g] \in K_j(S)$, then

$$[f] \star' [g] = [f] \star ((-1)^j [g]) = (-1)^j ([f] \star [g]) \quad (2.2.10)$$

In particular, for $\{u\} \in K_1(R)$ and $\{v\} \in K_1(S)$, the $\star'$ -product is given by the Steinberg symbol

$$\{u\} \star' \{v\} = \{u, v\}_{\text{St}}. \quad (2.2.11)$$

Proof. We consider the following commutative diagram

$$\begin{array}{ccc}
 K_i(R) \otimes K_j(S) & \xrightarrow{\star'} & K_{i+j}(R \otimes S) \\
 \downarrow \cong & & \downarrow \cong \\
 (-\star \{\tau\}) \otimes (-\star \{\tau\}) & & (-\star \{\tau\}) \star \{\tau\} \\
 \downarrow & & \downarrow \\
 K_{i+1}(\Sigma R) \otimes K_{j+1}(\Sigma S) & \xrightarrow{\star} & K_{i+j+2}(\Sigma^2(R \otimes S))
 \end{array} \tag{2.2.12}$$

induced by [Diagram \(2.2.7\)](#).

If $[f] \in K_i(R)$ and $[g] \in K_j(S)$, then the lower part of the diagram gives

$$\begin{aligned}
 ([f] \star \{\tau\}) \star ([g] \star \{\tau\}) &= [f] \star (\{\tau\} \star [g]) \star \{\tau\} && \text{(associativity)} \\
 &= [f] \star ((-1)^j ([g] \star \{\tau\})) \star \{\tau\} && \text{(graded-commutativity)} \\
 &= ([f] \star ((-1)^j [g]) \star \{\tau\}) \star \{\tau\}. && \text{(associativity)}
 \end{aligned}$$

So we must have

$$[f] \star' [g] = [f] \star ((-1)^j [g]).$$

□

2.3. Properties of the Extended Loday Pairing.

2.3.1. Relationship With the Classical Pairings in Algebraic K-Theory.

Milnor defined multiplicative structures on lower K-groups

$$\# : K_i(R) \otimes K_j(R) \rightarrow K_{i+j}(R \otimes R') \tag{2.3.1}$$

for $i, j \geq 0$, and $i + j \leq 2$ in [\[Mil72\]](#). Actually, Milnor only defined the multiplication internally (i.e., when $R = R'$, and is commutative). But one can mimic his definition to obtain an external product as in [Equation \(2.3.1\)](#), so that when $R = R'$ it becomes Milnor's version.

The multiplication in [Equation \(2.3.1\)](#) is given by the formulas:

$$\left\{ \begin{array}{ll} [P] \# [Q] &= [P \otimes Q] && \text{if } i = j = 0, \\ [P] \# \{u\} &= \{p \otimes u + (1 - p) \otimes 1\} && \text{if } i = 0, \text{ and } j = 1, \\ \{u\} \# \{v\} &= \{u, v\}_{\text{St}} && \text{if } i = j = 1. \end{array} \right. \tag{2.3.2}$$

Here, p is the idempotent operator associated to the projective module P . We omit the case $i = 0, j = 2$ here. The main tool we need is

Proposition 2.8 ([\[Mil72, Lemma 8.9 on page 70\]](#))

The multiplication $\#$ in [Equation \(2.3.1\)](#) is associative and bilinear for $i, j \geq 0$, and $i + j \leq 2$.

Note that Milnor's proofs extend to the external product case. It is also straight forward to verify that $\#$ is graded-commutative. We now relate $\#$ and $\star'$.

Theorem 2.9

For every $[P] \in K_0(R)$, $[Q] \in K_0(S)$, $\{u\} \in K_1(R)$, and $\{v\} \in K_1(S)$, we have

- (i) $[P] \star' [Q] = [P] \# [Q]$,
- (ii) $[P] \star' \{v\} = [P] \# (-\{v\}) = -([P] \# \{v\})$,
- (iii) $\{u\} \star' \{v\} = \{u\} \# \{v\}$.

Proof. We refer readers to [Equation \(2.2.5\)](#), and the commutative diagram in [Equation \(2.2.12\)](#).

(i) When $i = j = 0$, [Diagram \(2.2.7\)](#) induces the following commutative diagram

$$\begin{array}{ccc}
 K_0(R) \otimes K_0(R') & \xrightarrow{\star'} & K_0(R \otimes R') \\
 (-\# \{\tau\}) \otimes (-\# \{\tau\}) \downarrow \cong & & \cong \downarrow ((-\# \{\tau\}) \star \{\tau\}) \\
 K_1(\Sigma R) \otimes K_1(\Sigma R') & \xrightarrow{\star} & K_2(\Sigma^2(R \otimes R')).
 \end{array} \tag{2.3.3}$$

on homotopy. We then compute

$$\begin{aligned}
 & ([P] \# \{\tau\}) \star ([Q] \# \{\tau\}) \\
 &= -\{[P] \# \{\tau\}, [Q] \# \{\tau\}\}_{\text{St}} && \text{(Equation (2.1.11))} \\
 &= \{[P] \# \{\tau\}, (-[Q]) \# \{\tau\}\}_{\text{St}} && \text{(Bilinearity of } \{-, -\}_{\text{St}} \text{)} \\
 &= ([P] \# \{\tau\}) \# ((-[Q]) \# \{\tau\}) && \text{(Equation (2.3.2))} \\
 &= [P] \# (\{\tau\} \# (-[Q])) \# \{\tau\} && \text{(Associativity of } \# \text{)} \\
 &= [P] \# (-[Q] \# \{\tau\}) \# \{\tau\} && \text{(Graded-commutativity of } \# \text{)} \\
 &= -([P] \# ([Q] \# \{\tau\}) \# \{\tau\}) && \text{(Bilinearity of } \# \text{)} \\
 &= -\{[P] \# ([Q] \# \{\tau\}), \tau\}_{\text{St}} && \text{(Equation (2.3.2))} \\
 &= ([P] \# ([Q] \# \{\tau\})) \star \{\tau\} && \text{(Equation (2.1.11))}
 \end{aligned}$$

$$= ([P] \# [Q]) \# \{\tau\} \star \{\tau\}, \quad (\text{Associativity of } \#)$$

proving

$$[P] \star' [Q] = [P] \# [Q].$$

(ii) When $i = 0, j = 1$, [Diagram \(2.2.7\)](#) induces the following commutative diagram

$$\begin{array}{ccc} K_0(R) \otimes K_1(R') & \xrightarrow{\star'} & K_1(R \otimes R') \\ (-\# \{\tau\}) \otimes (-\star \{\tau\}) \cong & & \cong ((-\star \{\tau\}) \star \{\tau\}) \\ \downarrow & & \downarrow \\ K_1(\Sigma R) \otimes K_2(\Sigma R') & \xrightarrow{\star} & K_3(\Sigma^2(R \otimes R')). \end{array} \quad (2.3.4)$$

on homotopy groups. We then compute

$$\begin{aligned} & ([P] \# \{\tau\}) \star (\{v\} \star \{\tau\}) \\ &= (([P] \# \{\tau\}) \star \{v\}) \star \{\tau\} && (\text{Associativity of } \star) \\ &= (-\{[P] \# \{\tau\}, v\}_{\text{St}}) \star \{\tau\} && (\text{Proposition 2.4}) \\ &= (-((([P] \# \{\tau\}) \# \{v\}))) \star \{\tau\} && (\text{Equation (2.3.2)}) \\ &= (-([P] \# (\{\tau\} \# \{v\}))) \star \{\tau\} && (\text{Associativity of } \#) \\ &= ([P] \# (\{v\} \# \{\tau\})) \star \tau && (\text{Graded-commutativity of } \#) \\ &= (([P] \# \{v\}) \# \{\tau\}) \star \{\tau\} && (\text{Associativity of } \#) \\ &= \{[P] \# \{v\}, \{\tau\}\}_{\text{St}} \star \{\tau\} && (\text{Equation (2.3.2)}) \\ &= (-((([P] \# \{v\}) \star \{\tau\}))) \star \{\tau\} && (\text{Proposition 2.4}) \\ &= ((-([P] \# \{v\}) \star \{\tau\})) \star \{\tau\} && (\text{Bilinearity of } \star) \\ &= (([P] \# (-\{v\})) \star \{\tau\}) \star \{\tau\} && (\text{Bilinearity of } \#), \end{aligned}$$

proving

$$\begin{aligned}
& [P] \star' \{v\} = [P] \# (-\{v\}). \\
\text{(iii)} \quad & \{u\} \star' \{v\} = \{u, v\}_{\text{St}} \quad (\text{Proposition 2.7}) \\
& = \{u\} \# \{v\}.
\end{aligned}$$

□

The following case is still unknown.

Question 2.10

Is the product map

$$\star' : K_0(R) \otimes K_2(S) \rightarrow K_2(R \otimes S) \quad (2.3.5)$$

compatible with classical product map

$$\# : K_0(R) \otimes K_2(S) \rightarrow K_2(R \otimes S). \quad (2.3.6)$$

defined by Milnor in [Mil72, page 51]?

2.3.2. *The Non-Connective Gersten-Wagoner Algebraic K-Theory Spectrum $\mathbb{K}_R^{\text{GW}}$.*

We relate our extended Loday pairing

$$\gamma'_{\text{Loday}} : K_R \wedge K_S \rightarrow K_{R \otimes S}$$

to the structure map

$$\mathbb{S}^1 \wedge K_R \rightarrow K_{\Sigma R}$$

of the Gersten-Wagoner spectrum $\mathbb{K}_R^{\text{GW}}$. This will be used later in proving our version of the Loday assembly is a map of spectra.

In [Lod76, page 341–343], Loday gave an explicit description of the Gersten-Wagoner delooping

$$K_R \xrightarrow{\cong} \Omega K_{\Sigma R},$$

so that the induced isomorphisms on homotopy groups in Equation (2.2.2) are given as in Equation (2.2.5). We will use our extended Loday pairing to define a map

$$\mathbb{S}^1 \wedge K_R \rightarrow K_{\Sigma R},$$

whose adjoint

$$K_R \rightarrow \Omega K_{\Sigma R}$$

will induce isomorphisms on homotopy groups, and hence is a homotopy equivalence by the Whitehead Theorem. We begin by thinking the circle $\mathbb{S}^1$ as the classifying space $B\langle t \rangle$ of the infinite cyclic group generated by t , and define the following group homomorphism:

Definition 2.11 (The map $\mathfrak{t}^+ : B \langle t \rangle \rightarrow K_{\Sigma\mathbb{Z}}$)

Let $\langle t \rangle$ be the infinite cyclic group generated by t , and $\tau \in GL(\Sigma\mathbb{Z})$ be the element defined in Equation (2.2.4). Define the group homomorphism

$$\begin{aligned} \langle t \rangle &\rightarrow GL(\Sigma\mathbb{Z}) \\ t &\mapsto \tau^{-1}. \end{aligned}$$

The induced map

$$B \langle t \rangle \rightarrow K_{\Sigma\mathbb{Z}} \tag{2.3.7}$$

is denoted by $\mathfrak{t}^+$.

Proposition 2.12 (Another Gersten-Wagoner Delooping)

Let R be a ring, and $\langle t \rangle$ be the infinite cyclic group generated by t . The adjoint of the composition

$$B \langle t \rangle \wedge K_R \xrightarrow{\mathfrak{t}^+ \wedge \text{id}} K_{\Sigma\mathbb{Z}} \wedge K_R \xrightarrow{\gamma'_{\text{Loday}}} K_{\Sigma R} \tag{2.3.8}$$

is a weak equivalence, and hence a homotopy equivalence.

Proof. A class $[f] \in K_i(R)$ is represented by a spheroid

$$f : \mathbb{S}^i \rightarrow K_R.$$

Suspending it yields the following composition

$$\mathbb{S}^1 \wedge \mathbb{S}^i \xrightarrow{\text{id} \wedge f} \mathbb{S}^1 \wedge K_R \xrightarrow{\mathfrak{t}^+ \wedge \text{id}} K_{\Sigma\mathbb{Z}} \wedge K_R \xrightarrow{\gamma'_{\text{Loday}}} K_{\Sigma R}, \tag{2.3.9}$$

which represents the element $\{\tau^{-1}\} \star' [f] \in K_{i+1}(\Sigma R)$. In particular, we have

$$\{\tau^{-1}\} \star' [f] = \begin{cases} [f] \star \{\tau\} & \text{if } i > 0 \quad (\text{Proposition 2.7}) \\ [f] \# \{\tau\} & \text{if } i = 0 \quad (\text{Theorem 2.9 (ii)}) \end{cases} \tag{2.3.10}$$

The map on homotopy groups

$$K_i(R) \rightarrow K_{i+1}(\Sigma R)$$

induced by the adjoint of (2.3.8) sends

$$[f] \mapsto [f] \star' \{\tau^{-1}\},$$

which is an isomorphism for $i = 0$ (resp. $i > 0$) by [Lod76, after Theorem 1.4.7 on page 328] (resp. [Lod76, Corollary 2.3.6 on page 345]). Therefore, Whitehead Theorem asserts the homotopy equivalence

$$K_R \simeq K_{\Sigma R}.$$

□

Consequently, we have the following Ω -spectrum

Definition 2.13 (The Non-connective Gersten-Wagoner Algebraic K-theory Spectrum $\mathbb{K}_R^{\text{GW}}$)

Let R be a ring. Define the spectrum $\mathbb{K}_R^{\text{GW}}$ by having n -th space as

$$(\mathbb{K}_R^{\text{GW}})_n := \begin{cases} K_{\Sigma^n R} & \text{if } n \geq 0, \\ \Omega^{-n} K_R & \text{if } n < 0, \end{cases} \quad (2.3.11)$$

with the convention that $\Sigma^0 R := R$. The structure maps

$$f_n : (\mathbb{K}_R^{\text{GW}})_n \rightarrow (\mathbb{K}_R^{\text{GW}})_{n+1} \quad (2.3.12)$$

are given by

$$f_n := \begin{cases} (2.3.8) & \text{if } n \geq 0, \\ \text{adjoint of } \text{id}_{(\mathbb{K}_R^{\text{GW}})_n} & \text{if } n < 0. \end{cases} \quad (2.3.13)$$

By construction, $\mathbb{K}_R^{\text{GW}}$ is an Ω -spectrum.

3. THE MAIN THEOREM

3.1. Definition of the Loday Assembly. We use the extended Loday pairing γ'_{Loday} to define the Loday assembly map:

Definition 3.1 (The Loday Assembly Map α_{Loday} in Reduced Case)

Let R be a ring, and G be a group. The **Loday assembly map** is the map of spectra

$$\alpha_{\text{Loday}} : BG \wedge \mathbb{K}_R^{\text{GW}} \rightarrow \mathbb{K}_{R[G]}^{\text{GW}}, \quad (3.1.1)$$

whose components are given by the composition of maps of spaces:

$$BG \wedge K_{\Sigma^m R} \xrightarrow{j^+ \wedge \text{id}} K_{\mathbb{Z}[G]} \wedge K_{\Sigma^m R} \xrightarrow{\gamma'_{\text{Loday}}} K_{\Sigma^m R[G]}. \quad (3.1.2)$$

Here, the map

$$j^+ : BG \rightarrow K_{\mathbb{Z}[G]} \quad (3.1.3)$$

is induced by the group homomorphism

$$j : G \rightarrow GL(\mathbb{Z}[G])$$

$$g \mapsto \begin{bmatrix} g & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}. \quad (3.1.4)$$

We need to justify our definition.

Proposition 3.2 (Cf [Lod76, Proposition 4.1.1 on page 356])

Let R be a ring, and G be a group. The Loday assembly map

$\alpha_{\text{Loday}} : BG \wedge \mathbb{K}_R^{\text{GW}} \rightarrow \mathbb{K}_{R[G]}^{\text{GW}}$

defined in [Definition 3.1](#) is a weak map of spectra.

Proof. We need to show the components of α_{Loday} , as defined in [Equation \(3.1.2\)](#), commute with the structure maps of $\mathbb{K}_R^{\text{GW}}$ as defined in [Equation \(2.3.8\)](#). Let $\langle t \rangle$ be the infinite cyclic group generated by t . By unwrapping the definitions of α_{Loday} and the structure maps given in [Proposition 2.12](#), the following diagram

$$\begin{array}{ccccccc}
 \mathbb{S}^1 \wedge (BG \wedge K_R) & \xrightarrow{\text{id} \wedge (j^+ \wedge \text{id})} & \mathbb{S}^1 \wedge (K_{\mathbb{Z}[G]} \wedge K_R) & \xrightarrow{\text{id} \wedge \gamma'_{\text{Loday}}} & \mathbb{S}^1 \wedge K_{R[G]} \\
 \parallel & & \parallel & & \parallel \\
 BG \wedge B \langle t \rangle \wedge K_R & \xrightarrow{j^+ \wedge \text{id} \wedge \text{id}} & K_{\mathbb{Z}[G]} \wedge B \langle t \rangle \wedge K_R & \xrightarrow[\cong]{\text{twist}} & B \langle t \rangle \wedge K_{\mathbb{Z}[G]} \wedge K_R & \xrightarrow{\text{id} \wedge \gamma'_{\text{Loday}}} & B \langle t \rangle \wedge K_{R[G]} \\
 \downarrow \text{id} \wedge t^+ \wedge \text{id} & & \downarrow \text{id} \wedge t^+ \wedge \text{id} & & \downarrow t^+ \wedge \text{id} \wedge \text{id} & & \downarrow t^+ \wedge \text{id} \\
 BG \wedge K_{\Sigma\mathbb{Z}} \wedge K_R & \xrightarrow{j^+ \wedge \text{id} \wedge \text{id}} & K_{\mathbb{Z}[G]} \wedge K_{\Sigma\mathbb{Z}} \wedge K_R & \xrightarrow[\cong]{\text{twist}} & K_{\Sigma\mathbb{Z}} \wedge K_{\mathbb{Z}[G]} \wedge K_R & \xrightarrow{\text{id} \wedge \gamma'_{\text{Loday}}} & K_{\Sigma\mathbb{Z}} \wedge K_{R[G]} \\
 \downarrow \text{id} \wedge \gamma'_{\text{Loday}} & & \downarrow \text{id} \wedge \gamma'_{\text{Loday}} & & \downarrow \gamma'_{\text{Loday}} \circ \gamma'_{\text{Loday}} & & \downarrow \gamma'_{\text{Loday}} \\
 BG \wedge K_{\Sigma R} & \xrightarrow{j^+ \wedge \text{id}} & K_{\mathbb{Z}[G]} \wedge K_{\Sigma R} & \xrightarrow{\gamma'_{\text{Loday}}} & K_{\Sigma R[G]} & \xrightarrow{\text{id}} & K_{\Sigma R[G]}
 \end{array} \tag{3.1.5}$$

commutes up to weak homotopy by [Corollary 2.6](#). Here, the columns are the structure maps, and the rows are components of the Loday assembly. $\square$

The homotopy groups $\pi_i(BG \wedge \mathbb{K}_R^{\text{GW}})$ of the spectrum $BG \wedge \mathbb{K}_R^{\text{GW}}$ is the reduced homology groups of the classifying space BG of G with coefficients in $\mathbb{K}_R^{\text{GW}}$. So the induced map

$$\pi_i(\alpha_{\text{Loday}}) : \pi_i(BG \wedge \mathbb{K}_R^{\text{GW}}) \rightarrow K_i(R[G])$$

approximates the algebraic K-theory of the group ring $R[G]$ by a reduced homology theory. We want to lift the domain to an unreduced homology theory. Recall for any (based) topological space X , there is a split cofibre sequence

$$\mathbb{S}^0 \xleftarrow{\quad} X_+ \longrightarrow X \tag{3.1.6}$$

of spaces, where $X_+ := X \sqcup \text{point}$ is the space obtained by adding a disjoint point to X . If $\mathbb{E}$ is a spectrum, then we have a split cofibre sequence

$$\mathbb{S}^0 \wedge \mathbb{E} \xleftarrow{\quad} X_+ \wedge \mathbb{E} \longrightarrow X \wedge \mathbb{E} \tag{3.1.7}$$

of spectra. Consequently, we have a splitting

$$\begin{aligned}
 X_+ \wedge \mathbb{E} &\simeq (X \wedge \mathbb{E}) \vee (\mathbb{S}^0 \wedge \mathbb{E}) \\
 &\cong (X \wedge \mathbb{E}) \vee \mathbb{E}
 \end{aligned} \tag{3.1.8}$$

of spectra. This allows us to extend the Loday assembly to the unreduced case.

Definition 3.3 (The Loday Assembly Map α_{Loday} in Unreduced Case)

Let R be a ring and G be a group. The canonical inclusion $R \rightarrow R[G]$ induces a map

$$i : \mathbb{K}_R^{\text{GW}} \rightarrow \mathbb{K}_{R[G]}^{\text{GW}} \quad (3.1.9)$$

of spectra. The Loday assembly in unreduced case is defined by

$$BG_+ \wedge \mathbb{K}_R^{\text{GW}} \simeq (BG \wedge \mathbb{K}_R^{\text{GW}}) \vee \mathbb{K}_R^{\text{GW}} \xrightarrow{(\text{Definiton 3.1}) \vee i} \mathbb{K}_{R[G]}^{\text{GW}}. \quad (3.1.10)$$

By abuse of terminology and notation, we call this map the Loday assembly and denote it by α_{Loday} as well.

3.2. A Subgroup of the Source of Loday Assembly. The source of the Loday assembly

$$\pi_i(\alpha_{\text{Loday}}) : \pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\text{GW}}) \rightarrow K_{i+1}(R[G]) \quad (3.2.1)$$

on homotopy groups can be computed by the **Atiyah-Hirzebruch spectral sequence**:

$$\begin{aligned} \text{AHSS}(BG)_{p,q}^2 &\cong H_p(BG; K_q(R)) \\ &\Rightarrow \pi_{p+q}(BG_+ \wedge \mathbb{K}_R^{\text{GW}}), \end{aligned} \quad (3.2.2)$$

and in particular,

$$\begin{aligned} E_{1,i}^2 &\cong H_1(BG; K_i(R)) \\ &\cong G_{ab} \otimes K_i(R). \end{aligned} \quad (3.2.3)$$

Because the Atiyah-Hirzebruch spectral sequence is concentrated on the right-half plane, it follows that the differential

$$d_{1,i}^2 : E_{1,i}^2 \rightarrow E_{-1,i+1}^2 \quad (3.2.4)$$

has to be zero. So the group $E_{1,i}^\infty$ is a quotient of $E_{1,i}^2$, and we shall see it is also a subgroup of the source $\pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\text{GW}})$.

Fix a spectrum $\mathbb{E}$. For any CW complex X , we have a homeomorphism

$$X_+ \cong X \vee \mathbb{S}^0, \quad (3.2.5)$$

and hence an isomorphism of homotopy groups

$$\pi_{i+1}(X_+ \wedge \mathbb{E}) \cong \pi_{i+1}(X \wedge \mathbb{E}) \oplus \pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E}). \quad (3.2.6)$$

Fix a skeleta filtration

$$\text{point} = X^{(0)} \subseteq X^{(1)} \subseteq \cdots \subseteq X \quad (3.2.7)$$

on X , the identification in Equation (3.2.5) gives a skeleta filtration

$$X_+^{(p)} \cong X^{(p)} \vee \mathbb{S}^0 \quad (3.2.8)$$

on X_+ . Thus we have a splitting for the Atiyah-Hirzebruch spectral sequence:

$$\text{AHSS}(X_+)^r_{p,q} \cong \text{AHSS}(X)^r_{p,q} \oplus \text{AHSS}(\mathbb{S}^0)^r_{p,q} \quad (3.2.9)$$

for $r \geq 1$. Tracing through the construction of the Atiyah-Hirzebruch spectral sequence (see, for example [Wic15]), we see that the splitting in Equation (3.2.5) respects the filtration on homotopy groups:

$$\mathcal{F}_1 [\pi_{i+1}(X_+ \wedge \mathbb{E})] \cong \mathcal{F}_1 [\pi_{i+1}(X \wedge \mathbb{E})] \oplus \mathcal{F}_1 [\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E})], \quad (3.2.10)$$

for which

$$\mathcal{F}_1 [\pi_{i+1}(X_+ \wedge \mathbb{E})] := \text{im} \left(\pi_{i+1} \left(X_+^{(1)} \wedge \mathbb{E} \right) \rightarrow \pi_{i+1}(X_+ \wedge \mathbb{E}) \right). \quad (3.2.11)$$

The subgroup

$$\mathcal{F}_1 [\pi_{i+1}(X_+ \wedge \mathbb{E})] \subseteq \pi_{i+1}(X_+ \wedge \mathbb{E}) \quad (3.2.12)$$

is what we are interested in, and it can be identified as:

$$\begin{aligned} \mathcal{F}_1 [\pi_{i+1}(X_+ \wedge \mathbb{E})] &\cong \mathcal{F}_1 [\pi_{i+1}(X \wedge \mathbb{E})] \oplus \mathcal{F}_1 [\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{E})] \\ &= \text{AHSS}(X)_{1,i}^\infty \oplus \pi_{i+1}(\mathbb{E}). \end{aligned} \quad (3.2.13)$$

3.3. Statement of the Main Theorem and the Proof. Now, let us work with $\mathbb{E} = \mathbb{K}_R^{\text{GW}}$, and $X = BG$. Then Equation (3.2.13) says

$$\begin{aligned} \mathcal{F}_1 [\pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\text{GW}})] &\cong \mathcal{F}_1 [\pi_{i+1}(BG \wedge \mathbb{K}_R^{\text{GW}})] \oplus \mathcal{F}_1 [\pi_{i+1}(\mathbb{S}^0 \wedge \mathbb{K}_R^{\text{GW}})] \\ &= \text{AHSS}(BG)_{1,i}^\infty \oplus \pi_{i+1}(\mathbb{K}_R^{\text{GW}}) \\ &= \text{AHSS}(BG)_{1,i}^\infty \oplus K_{i+1}(R). \end{aligned} \quad (3.3.1)$$

So the source of the Loday assembly

$$\alpha_{\text{Loday}} : \pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\text{GW}}) \rightarrow K_{i+1}(R[G])$$

contains

$$\text{AHSS}(BG)_{1,i}^\infty \oplus K_{i+1}(R)$$

as a subgroup. In fact, we already know what the Loday assembly is doing when restricted onto the summand $K_{i+1}(R)$ —it is induced by the inclusion

$$i : R \rightarrow R[G]$$

of rings. See [Lod76, before Lemma 4.1.2 on page 356], or Definition 3.3 before. What we want now is to study the Loday assembly when restricted onto the summand $\text{AHSS}(BG)_{1,i}^\infty$.

The splitting of the Atiyah-Hirzebruch spectral sequence in Equation (3.2.9) says the differentials of $\text{AHSS}(BG_+)_{p,q}^2$ splits into two parts, which we illustrate here:

$$\begin{array}{ccc}
\begin{array}{c} \uparrow q \\
\begin{array}{l}
0 \ H_1(BG; K_5(R)) H_2(BG; K_5(R)) \cdots \\
0 \ H_1(BG; K_4(R)) H_2(BG; K_4(R)) \cdots \\
0 \ H_1(BG; K_3(R)) H_2(BG; K_3(R)) \cdots \\
0 \ H_1(BG; K_2(R)) H_2(BG; K_2(R)) \cdots \\
0 \ H_1(BG; K_1(R)) H_2(BG; K_1(R)) \cdots \\
0 \ H_1(BG; K_0(R)) H_2(BG; K_0(R)) \cdots
\end{array} \\
\rightarrow p
\end{array} & \cong & \begin{array}{c} \uparrow q \\
\begin{array}{l}
0 \ H_1(BG; K_5(R)) \ H_2(BG; K_5(R)) \ \cdots \\
0 \ H_1(BG; K_4(R)) \ H_2(BG; K_4(R)) \ \cdots \\
0 \ H_1(BG; K_3(R)) \ H_2(BG; K_3(R)) \ \cdots \\
0 \ H_1(BG; K_2(R)) \ H_2(BG; K_2(R)) \ \cdots \\
0 \ H_1(BG; K_1(R)) \ H_2(BG; K_1(R)) \ \cdots \\
0 \ H_1(BG; K_0(R)) \ H_2(BG; K_0(R)) \ \cdots
\end{array} \\
\rightarrow p
\end{array} \oplus \begin{array}{c} \uparrow q \\
\begin{array}{l}
K_5(R) \ 0 \ 0 \cdots \\
K_4(R) \ 0 \ 0 \cdots \\
K_3(R) \ 0 \ 0 \cdots \\
K_2(R) \ 0 \ 0 \cdots \\
K_1(R) \ 0 \ 0 \cdots \\
K_0(R) \ 0 \ 0 \cdots
\end{array} \\
\rightarrow p
\end{array} \\
\text{AHSS}(BG)_{p,q}^2 & & \text{AHSS}(BG)_{p,q}^2 \oplus \text{AHSS}(\mathbb{S}^0)_{p,q}^2
\end{array} \tag{3.3.2}$$

(We omit the negative K-groups because they are annoying to include.) We note that the groups $\text{AHSS}(BG)_{0,q}^2$ are trivial because the reduced homology group $\widetilde{H}_0(BG)$ is trivial for BG being path-connected. Consequently,

Lemma 3.4

Let G be a group, and R be a ring. The differentials

$$d_{1,q}^2 : \text{AHSS}(BG_+)_{1,q}^2 \rightarrow \text{AHSS}(BG_+)_{-1,q+1}^2 \tag{3.3.3}$$

in the E^2 -page of the Atiyah-Hirzebruch spectral sequence of BG_+ associated to the K-theory spectrum $\mathbb{K}_R^{\text{GW}}$ are trivial for all q .

Proof. The splitting of the Atiyah-Hirzebruch spectral sequence in Equation (3.2.9) says the differential

$$d_{1,q}^2 : \text{AHSS}(BG_+)_{1,q}^2 \rightarrow \text{AHSS}(BG_+)_{-1,q+1}^2$$

splits into direct sum of the two differentials

$$\text{AHSS}(BG)_{1,q}^2 \rightarrow \text{AHSS}(BG)_{-1,q+1}^2, \tag{3.3.4}$$

$$\text{AHSS}(\mathbb{S}^0)_{1,q}^2 \rightarrow \text{AHSS}(\mathbb{S}^0)_{-1,q+1}^2. \tag{3.3.5}$$

From Diagram (3.3.2), we see that the first (resp. second) map is trivial because the codomain (resp. domain) is trivial, hence proving the claim. $\square$

Lemma 3.4 says the group $\text{AHSS}(BG_+)_{1,i}^\infty$ is a quotient of $\text{AHSS}(BG_+)_{1,i}^2$, and therefore we have a diagram

$$\begin{array}{ccc}
\text{AHSS}(BG)_{1,i}^2 \cong G_{ab} \otimes K_i(R) & \xrightarrow{\Phi_i} & K_{i+1}(R[G]) \\
\downarrow & & \uparrow \pi_{i+1}(\alpha_{\text{Loday}}) \\
\text{AHSS}(BG)_{1,i}^\infty & \hookrightarrow & \pi_{i+1}(BG \wedge \mathbb{K}_R^{\text{GW}}).
\end{array} \tag{3.3.6}$$

We would like to study the filler Φ_i . (Perhaps we should explain the absent of the disjoint base-point “+” in [Diagram \(3.3.6\)](#) is not a typo. As pointed out in [Equation \(3.3.1\)](#), $\pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\text{GW}})$ contains $K_{i+1}(R) \oplus \text{AHSS}(BG)_{1,i}^\infty$ as a subgroup, and we want to know what the Loday assembly is doing on the term $\text{AHSS}(BG)_{1,i}^\infty$.)

Let us go back to the E^1 -page of the Atiyah-Hirzebruch spectral sequence. Unlike the later pages, the E^1 -page is not homotopy invariant—it depends on the choice of cellular structure on BG . **For our purpose, we are using the skeletal filtration on the bar construction for BG ,** so that

$$\text{AHSS}(BG)_{1,i}^1 \cong \pi_{i+1} \left(\left(\bigvee_{g \in G} \mathbb{S}^1 \right) \wedge \mathbb{K}_R^{\text{GW}} \right). \tag{3.3.7}$$

The filler we want to study is then induced by the composition

$$\pi_{i+1} \left(\left(\bigvee_{g \in G} \mathbb{S}^1 \right) \wedge \mathbb{K}_R^{\text{GW}} \right) \xrightarrow{(i_G)_*} \pi_{i+1}(BG \wedge \mathbb{K}_R^{\text{GW}}) \xrightarrow{\pi_{i+1}(\alpha_{\text{Loday}})} K_{i+1}(R[G]), \tag{3.3.8}$$

for which the first arrow is induced by the inclusion

$$i_G : \bigvee_{g \in G} \mathbb{S}^1 \hookrightarrow BG \tag{3.3.9}$$

of the 1-skeleton into BG . This follows from the construction of the Atiyah-Hirzebruch spectral sequence. (See [\[Wic15\]](#) for example.) When $G = \langle t \rangle$ is the infinite cyclic group, we know the circle $\mathbb{S}^1$ is a model for $B\langle t \rangle$. Moreover,

Lemma 3.5

The inclusion

$$i_t : \mathbb{S}^1 \rightarrow B\langle t \rangle \tag{3.3.10}$$

that sends $\mathbb{S}^1$ to the 1-cell of $B\langle t \rangle$ labelled by the generator t is a weak equivalence.

Proof. It is clear that the induced group homomorphism

$$(i_t)_* : \pi_i(\mathbb{S}^1) \rightarrow \pi_i(B\langle t \rangle) \tag{3.3.11}$$

is an isomorphism for $i \neq 1$, because the source and target are both trivial. So we need to check the map

$$(i_t)_* : \pi_1(\mathbb{S}^1) \rightarrow \pi_1(B\langle t \rangle) \tag{3.3.12}$$

is an isomorphism.

The map in Equation (3.3.12) is just an endomorphism of the infinite cyclic group. In particular, it is surjective by the definition of $\mathbf{i}_t$. (The generator is in the image.) So the First Isomorphism Theorem says the map in Equation (3.3.12) must have trivial kernel, and hence an isomorphism. $\square$

The Whitehead Theorem then asserts $\mathbf{i}_t$ is a homotopy equivalence. Thus when $G = \langle t \rangle$ is the infinite cyclic group, we have the following commutative diagram

$$\begin{array}{ccc}
 \bigvee_{g \in \langle t \rangle} \mathbb{S}^1 & \xrightarrow{\mathbf{i}_{\langle t \rangle}} & B \langle t \rangle \\
 & \nwarrow \mathbf{j}_t & \uparrow \simeq \mathbf{i}_t \\
 & & \mathbb{S}^1
 \end{array} \tag{3.3.13}$$

The map $\mathbf{j}_t$ sends $\mathbb{S}^1$ to the loop in $\bigvee_{g \in \langle t \rangle} \mathbb{S}^1$ labelled by the generated t . The upshot is that we are able to describe the composition in Equation (3.3.8) when $G = \langle t \rangle$.

Proposition 3.6

When $G = \langle t \rangle$ is the infinite cyclic group, the composition in Equation (3.3.8) can be identified as

$$\begin{array}{ccccc}
 \pi_{i+1} \left(\left(\bigvee_{g \in \langle t \rangle} \mathbb{S}^1 \right) \wedge \mathbb{K}_R^{\text{GW}} \right) & \xrightarrow{(\mathbf{i}_{\langle t \rangle})} & \pi_{i+1} (B \langle t \rangle \wedge \mathbb{K}_R^{\text{GW}}) & \xrightarrow{\pi_{i+1} (\alpha_{\text{Loday}})} & K_{i+1} (R[t^{\pm 1}]) \\
 \uparrow (\mathbf{j}_t)_* & & \uparrow (\mathbf{i}_t)_* \cong & & \parallel \\
 \pi_{i+1} (\mathbb{S}^1 \wedge \mathbb{K}_R^{\text{GW}}) & \xrightarrow{\text{id}} & \pi_{i+1} (\mathbb{S}^1 \wedge \mathbb{K}_R^{\text{GW}}) & \longrightarrow & K_{i+1} (R[t^{\pm 1}]) \\
 \parallel & & \parallel & & \parallel \\
 K_i(R) & \xrightarrow{\text{id}} & K_i(R) & \longrightarrow & K_{i+1} (R[t^{\pm 1}]),
 \end{array} \tag{3.3.14}$$

where the bottom composition sends $[f] \in K_i(R)$ to

$$[f] \star \{t^{-1}\} \in K_{i+1} (R[t^{\pm 1}]),$$

We remind readers the commutativity of the top-left square comes from Diagram (3.3.13).

Proof. The class $[f] \in K_i(R)$ is represented by the spheroid

$$f : \mathbb{S}^i \rightarrow K_R,$$

where

$$K_R := K_0(R) \times BGL(R)^+.$$

Taking suspension yields

$$\mathbb{S}^1 \wedge \mathbb{S}^i \xrightarrow{\text{id} \wedge f} \mathbb{S}^1 \wedge K_R \xrightarrow{j^+ \wedge \text{id}} K_{\mathbb{Z}[t^{\pm 1}]} \wedge K_R \xrightarrow{\gamma'_{\text{Loday}}} K_{R[t^{\pm 1}]}.$$

This composition represents the element $\pi_{i+1}(\alpha_{\text{Loday}})([f]) \in K_{i+1}(R[t^{\pm 1}])$, as well as the product

$$[f] \star' \{j^+\} \in K_{i+1}(R[t^{\pm 1}]).$$

Now, j^+ is induced by the group homomorphism

$$j : \langle t \rangle \rightarrow GL(\mathbb{Z}[t^{\pm 1}])$$

$$t \mapsto \begin{bmatrix} t & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}.$$

So $\{j^+\} = \{t\}$, and thus

$$\begin{aligned} [f] \star' \{j^+\} &= [f] \star' \{t\} \\ &= [f] \star (-\{t\}) \\ &= [f] \star \{t^{-1}\} \end{aligned} \quad (\text{Proposition 2.7})$$

as desired. □

For arbitrary group G , we recall the isomorphism

$$\begin{aligned} G &\cong \text{Hom}(\langle t \rangle, G) \\ g &\mapsto (\varphi_g : t \mapsto g), \end{aligned} \quad (3.3.15)$$

allowing us to extend [Proposition 3.6](#) to arbitrary G .

Proposition 3.7

For an arbitrary group G , under the canonical isomorphism

$$\pi_{i+1} \left(\left(\bigvee_{g \in G} \mathbb{S}^1 \right) \wedge \mathbb{K}_R^{\text{GW}} \right) \cong \bigoplus_{g \in G} K_i(R), \quad (3.3.16)$$

the composition in (3.3.8) can be identified as

$$\begin{array}{ccccc}
\pi_{i+1} \left(\left(\bigvee_{g \in G} \mathbb{S}^1 \right) \wedge \mathbb{K}_R^{\text{GW}} \right) & \xrightarrow{(\mathbf{i}_G)_*} & \pi_{i+1} (BG \wedge \mathbb{K}_R^{\text{GW}}) & \xrightarrow{\pi_{i+1}(\alpha_{\text{Loday}})} & K_{i+1}(R[G]) \\
\parallel & & \parallel & & \parallel \\
\bigoplus_{g \in G} K_i(R) & \longrightarrow & \pi_{i+1} (BG \wedge \mathbb{K}_R^{\text{GW}}) & \longrightarrow & K_{i+1}(R[G]),
\end{array} \tag{3.3.17}$$

where the bottom composition sends the element $[f] \in K_i(R)$ in the summand in $\bigoplus_{g \in G} K_i(R)$ labelled by $g \in G$ to the element

$$[f] \star \{g^{-1}\} \in K_{i+1}(R[G]).$$

Proof. Fix $g \in G$, and consider the commutative diagram

$$\begin{array}{ccc}
\pi_{i+1}(\mathbb{S}^1 \wedge \mathbb{K}_R^{\text{GW}}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1} \left(\left(\bigvee_{g \in G} \mathbb{S}^1 \right) \wedge \mathbb{K}_R^{\text{GW}} \right) \\
\downarrow (\mathbf{i}_t)_* \cong & & \downarrow (\mathbf{i}_G)_* \\
\pi_{i+1}(B\langle t \rangle \wedge \mathbb{K}_R^{\text{GW}}) & \xrightarrow{(\varphi_g)_*} & \pi_{i+1}(BG \wedge \mathbb{K}_R^{\text{GW}}) \\
\downarrow \pi_{i+1}(\alpha_{\text{Loday}}) & & \downarrow \pi_{i+1}(\alpha_{\text{Loday}}) \\
K_{i+1}(R[t^{\pm 1}]) & \xrightarrow{(\varphi_g)_*} & K_{i+1}(R[G]).
\end{array} \tag{3.3.18}$$

The top square commutes by chasing the definitions of the arrows. The bottom square commutes by the naturality of α_{Loday} via the group homomorphism

$$\varphi_g : \langle t \rangle \rightarrow G.$$

Now, [Proposition 3.6](#) says the left vertical composition in [Diagram \(3.3.18\)](#) sends $[f] \in K_i(R)$ to

$$[f] \star \{t^{-1}\} \in K_{i+1}(R[t^{\pm 1}]).$$

So the commutativity of [Diagram \(3.3.18\)](#) says the right vertical composition sends $[f] \in K_i(R)$ in the summand $\bigoplus_{g \in G} K_i(R)$ indexed by $g \in G$ to the element

$$[f] \star \{g^{-1}\} \in K_{i+1}(R[G]).$$

□

Theorem 3.8 (An Explicit Formula for the Loday Assembly)

The filler Φ_i in [Diagram \(3.3.6\)](#) is induced by the group homomorphism

$$\begin{aligned} G \times K_i(R) &\rightarrow K_{i+1}(R[G]) \\ (g, [f]) &\mapsto [f] \star \{g^{-1}\}. \end{aligned} \quad (3.3.19)$$

Proof. Follows immediately from [Proposition 3.7](#). □

Corollary 3.9 (The Loday Assembly on π_1 , [[Wal78b](#), Assertion 15.8 on page 229], [[LR05](#), page 708])

For a regular ring R , and a group G , the Loday assembly on π_1 is given by

$$K_1(i) \oplus \Phi_1 : K_1(R) \oplus [G_{ab} \otimes K_0(R)] \rightarrow K_1(R[G]), \quad (3.3.20)$$

for which $i : R \rightarrow R[G]$ is the inclusion, and Φ_1 is induced by the map

$$\begin{aligned} G \times K_0(R) &\rightarrow K_1(R[G]) \\ (g, [P]) &\mapsto \{\mathfrak{h}_g\}, \end{aligned} \quad (3.3.21)$$

for which $\mathfrak{h}_g : P \otimes_R R[G] \rightarrow R \otimes_R R[G]$ is the automorphism given by

$$\mathfrak{h}_g(x \otimes u) := x \otimes ug^{-1}.$$

Proof. If the ring R is regular, then the Atiyah-Hirzebruch spectral sequence for $\pi_i(BG_+ \wedge \mathbb{K}_R^{\text{GW}})$ is first-quadrant. Hence,

$$\pi_1(BG_+ \wedge \mathbb{K}_R^{\text{GW}}) \cong K_1(R) \oplus [G_{ab} \otimes K_0(R)].$$

If $g \in G$, and $[P] \in K_0(R)$, then [Theorem 3.8](#) says the Loday assembly sends $(g, [P])$ to

$$\begin{aligned} [P] \star \{g^{-1}\} &= [P] \star' \{g\} \\ &= [P] \# \{g^{-1}\} && \text{(Theorem 2.9)} \\ &= \{p \otimes g^{-1} + (1 - p) \otimes 1\}. && \text{(Equation (2.3.2))} \end{aligned}$$

If $x \otimes u \in P \otimes_R R[G]$, then

$$\begin{aligned} (p \otimes g^{-1} + (1 - p) \otimes 1)(x \otimes u) &= (p \otimes g^{-1})(x \otimes u) + ((1 - p) \otimes 1)(x \otimes u) \\ &= (p \otimes g^{-1})(x \otimes u) \\ &\quad (\text{since } (1 - p)(x) = 0) \\ &= x \otimes ug^{-1}. \end{aligned}$$

Therefore,

$$p \otimes g^{-1} + (1 - p) \otimes 1 = \mathfrak{h}_g.$$

□

Corollary 3.10 ([Wal78b, Proposition 15.7 (1) on page 229])

Let G be a group. The cokernel of the Loday assembly

$$\text{coker}(\pi_1(\alpha_{\text{Loday}}) : \pi_1(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \rightarrow K_1(\mathbb{Z}[G])) \quad (3.3.22)$$

on π_1 is isomorphic to the first Whitehead group

$$\text{Wh}_1(G) := \frac{K_1(\mathbb{Z}[G])}{\pm G}. \quad (3.3.23)$$

Proof. We need to check the image $\text{im}(\pi_1(\alpha_{\text{Loday}}))$ of the Loday assembly on π_1 is the subgroup $\pm G_{ab}$ of $K_1(\mathbb{Z}[G])$.

Note that the ring $\mathbb{Z}$ has no negative K-groups for being regular. Thus, the vanishing of the differentials of the Atiyah-Hirzebruch spectral sequences says

$$\pi_1(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \cong K_1(\mathbb{Z}) \oplus [G_{ab} \otimes K_0(\mathbb{Z})] \quad (3.3.24)$$

$$\cong C_2 \oplus [G_{ab} \otimes \mathbb{Z}]$$

$$\cong \{1, -1\} \oplus G_{ab} \quad (3.3.25)$$

We already know, from Definition 3.3, the Loday assembly $\pi_1(\alpha_{\text{Loday}})|_{K_1(\mathbb{Z})}$ when restricted onto the summand $K_1(\mathbb{Z})$ of $\pi_1(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}})$ is the group homomorphism

$$K_1(\mathbb{Z}) \rightarrow K_1(\mathbb{Z}[G])$$

induced by the canonical inclusion $i : \mathbb{Z} \rightarrow \mathbb{Z}[G]$. Therefore, under the identification

$$K_1(\mathbb{Z}) \cong \{1, -1\},$$

the restriction $\pi_1(\alpha_{\text{Loday}})|_{K_1(\mathbb{Z})}$ sends ± 1 to $\pm 1 \in K_1(\mathbb{Z}[G])$.

On the other hand, we observe that the map

$$\Phi_1 : G_{ab} \otimes K_0(\mathbb{Z}) \rightarrow K_1(\mathbb{Z}[G])$$

from Corollary 3.9 sends the simple tensor $g \otimes [\mathbb{Z}] \in G_{ab} \otimes K_0(\mathbb{Z})$ to the element $\{g^{-1}\} \in K_1(\mathbb{Z}[G])$. Property of group homomorphism then tells us that Φ_1 sends the simple tensor $g \otimes [\mathbb{Z}^n] \in G_{ab} \otimes K_0(\mathbb{Z})$ to the element $\{g^{-n}\} \in K_1(\mathbb{Z}[G])$ represented by multiplication by g^{-n} . As a result, under the identification

$$G_{ab} \otimes K_0(\mathbb{Z}) \cong G_{ab},$$

the map Φ_1 sends $g \in G_{ab}$ to $\{g^{-1}\} \in K_1(\mathbb{Z}[G])$.

Combining everything together, we see that, under the identification

$$\pi_1(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \cong \{\pm 1\} \oplus G_{ab},$$

the Loday assembly $\pi_1(\alpha_{\text{Loday}})$ sends the element $(\pm 1) \oplus g \in \{\pm 1\} \oplus G_{ab}$ to the element $\{\pm g^{-1}\} \in K_1(\mathbb{Z}[G])$. This proves $\text{im}(\pi_1(\alpha_{\text{Loday}}))$ is the subgroup $\pm G$ of $K_1(\mathbb{Z}[G])$ as desired. $\square$

4. THE NON-INJECTIVITY PROBLEM FOR THE INTEGRAL LODAY ASSEMBLY

One might ask whether torsion-free is a necessary condition in the [Classical Farrell-Jones Conjecture 1.3](#). The point is that the cokernel

$$\text{coker}(\pi_i(\alpha_{\text{Loday}}) : \pi_i(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \rightarrow K_i(\mathbb{Z}[G])) \cong \text{Wh}_i(G) \quad (4.0.1)$$

can be identified with the classical Whitehead groups $\text{Wh}_i(G)$ for $i = 0, 1$ and 2 . See [\[Lod76, page 357–364\]](#), or [Corollary 3.10](#) for the case $i = 1$. So, non-vanishing $\text{Wh}_i(G)$ implies the non-surjectivity of $\pi_i(\alpha_{\text{Loday}})$. For example, if p is an odd prime and C_p is the cyclic group of order p , then $\text{Wh}_1(C_p) \neq 0$ [\[Coh73, 11.5 on page 45\]](#). Therefore, torsion-free is necessary to guarantee surjectivity. What about injectivity?

Question 4.1 (Non-injectivity Problem for the Loday Assembly)

Let R be a regular ring. Is there a group G with torsion, such that Loday assembly

$$\pi_i(\alpha_{\text{Loday}}) : \pi_i(BG_+ \wedge \mathbb{K}_R^{\text{GW}}) \rightarrow K_i(R[G])$$

is not injective for some i ?

It turns out that under suitable conditions, the Loday assembly is injective in lower degrees.

Proposition 4.2 ([LR05, Lemma 2 on page 709])

Let R be a regular ring and G be a group.

(i) The Loday assembly

$$\pi_0(\alpha_{\text{Loday}}) : \pi_0(BG_+ \wedge \mathbb{K}_R^{\text{GW}}) \rightarrow K_0(R[G])$$

on π_0 is always injective. In particular, its left inverse is induced by the augmentation map

$$R[G] \rightarrow R.$$

(ii) If also R is commutative, and the natural map

$$\begin{aligned} \mathbb{Z} &\rightarrow K_0(R) \\ 1 &\mapsto [R] \end{aligned}$$

is an isomorphism, then the Loday assembly

$$\pi_1(\alpha_{\text{Loday}}) : \pi_1(BG_+ \wedge \mathbb{K}_R^{\text{GW}}) \rightarrow K_1(R[G])$$

on π_1 is injective.

Consequently, the first place to look for non-injectivity phenomena for Loday assembly would be the second homotopy group.

4.1. Second Homotopy Group. [Question 4.1](#) was answered by Ullmann-Wu in the case when R is a finite field.

Theorem 4.3 ([UW17, Theorem 2 on page 461])

Let G be a finite group such that $H_2(G; \mathbb{Z})$ is non-trivial, and $\mathbb{F}$ a finite field with characteristic p which does not divide the order of G , then the Loday assembly

$$\pi_2(\alpha_{\text{Loday}}) : \pi_2(BG_+ \wedge \mathbb{K}_{\mathbb{F}}^{\text{GW}}) \rightarrow K_2(\mathbb{F}[G])$$

is not injective.

For example, we can take $G = C_2 \oplus C_2$ and $\mathbb{F}$ to be any finite field with characteristic $p > 2$.

However, no such example is known when $R = \mathbb{Z}$. In this case, the Atiyah-Hirzebruch spectral sequence for $\pi_2(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}})$ yields the following short exact sequence

$$0 \rightarrow K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \rightarrow \pi_2(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \rightarrow H_2(BG; K_0(\mathbb{Z})) \rightarrow 0. \quad (4.1.1)$$

See [Leh18, Theorem 12.2 on page 29]. Together with Theorem 3.8, we have the following corollary:

Corollary 4.4 (The Loday Assembly on π_2 of an Integral Group Ring)

Let G be a group. The Loday assembly map

$$\alpha_{\text{Loday}} : BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}} \rightarrow \mathbb{K}_{\mathbb{Z}[G]}^{\text{GW}} \quad (4.1.2)$$

for the integral group ring $\mathbb{Z}[G]$ on π_2 , when restricted onto the subgroup

$$K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \leq \pi_2(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}),$$

is given by the formula:

$$K_2(i) \oplus \Phi_2 : K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})] \rightarrow K_2(\mathbb{Z}[G]), \quad (4.1.3)$$

where $i : \mathbb{Z} \rightarrow \mathbb{Z}[G]$ is the inclusion, and Φ_2 is induced by the map

$$\begin{aligned} G \times K_1(\mathbb{Z}) &\rightarrow K_2(\mathbb{Z}[G]) \\ (g, \pm 1) &\mapsto \{\pm 1, g\}_{\text{St}}. \end{aligned} \quad (4.1.4)$$

In particular, the Steinberg symbol $\{1, g\}_{\text{St}}$ is always the identity element in $K_2(\mathbb{Z}[G])$ for all $g \in G$.

Proof. Only the description of Φ_2 needs to be justified. Theorem 3.8 says Φ_2 is induced by the map

$$\begin{aligned} G \times K_1(\mathbb{Z}) &\rightarrow K_2(\mathbb{Z}[G]) \\ (g, \pm 1) &\mapsto \{\pm 1\} \star \{g^{-1}\}, \end{aligned} \quad (4.1.5)$$

and we have

$$\begin{aligned} \{\pm 1\} \star \{g^{-1}\} &= -\{\pm 1, g^{-1}\}_{\text{St}} && \text{(Proposition 2.4)} \\ &= \{\pm 1, (g^{-1})^{-1}\}_{\text{St}} && \text{(Bilinearity of the Steinberg symbol)} \\ &= \{\pm 1, g\}_{\text{St}} \end{aligned}$$

as desired. □

It turns out that injectivity can happen for group with torsion.

Example 4.5 (Bijectivity on $K_2(\mathbb{Z}[C_2])$) Denote by C_2 the cyclic group of order two, with generator g . Dunwoody showed that the second K-group of $\mathbb{Z}[C_2]$ is generated by two Steinberg symbols [Dun75, Theorem on page 482]:

$$\begin{aligned} K_2(\mathbb{Z}[C_2]) &= \langle \{-1, -1\}_{\text{St}}, \{-1, g\}_{\text{St}} \rangle \\ &\cong C_2 \oplus C_2. \end{aligned} \tag{4.1.6}$$

Because

$$\begin{aligned} H_2(BC_2; K_0(\mathbb{Z})) &\cong H_2(BC_2; \mathbb{Z}) \\ &= 0, \end{aligned}$$

we have

$$\pi_2(BC_{2+} \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \cong K_2(\mathbb{Z}) \oplus [(C_2)_{ab} \otimes K_1(\mathbb{Z})]$$

Corollary 4.4 then says the Loday assembly for C_2

$$\pi_2(\alpha_{\text{Loday}}) = K_2(i) \oplus \Phi_2 \tag{4.1.7}$$

is bijective on π_2 .

The integral case of Question 4.1 on π_2 is then related to deciding whether the Steinberg symbol $\{-1, g\}_{\text{St}} \in K_2(\mathbb{Z}[G])$ is trivial or not. In particular, we have

Theorem 4.6 (An Injectivity Result for the Loday Assembly on π_2)

Let G be a group such that the second homology group $H_2(BG; \mathbb{Z})$ is trivial. Then the Loday assembly

$$\pi_2(\alpha_{\text{Loday}}) : \pi_2(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \rightarrow K_i(\mathbb{Z}[G])$$

on π_2 is injective if and only if the identity in G is the only element that makes the Steinberg symbol $\{-1, g\}_{\text{St}} \in K_2(\mathbb{Z}[G])$ trivial.

Proof. If $H_2(BG; \mathbb{Z})$ is trivial, then the short exact sequence in Equation (4.1.1) says

$$\pi_2(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}}) \cong K_2(\mathbb{Z}) \oplus [G_{ab} \otimes K_1(\mathbb{Z})].$$

Corollary 4.4 then tells us that the Loday assembly on π_2 is given by

$$\pi_2(\alpha_{\text{Loday}}) = K_2(i) \oplus \Phi_2.$$

Now, the map $K_2(i)$ is always injective, so the injectivity/non-injectivity behaviour of $\pi_2(\alpha_{\text{Loday}})$ is completely determined by Φ_2 in this case. Again, Corollary 4.4 says Φ_2 is injective if and only if the identity in G is the only element that makes the Steinberg symbol $\{-1, g\}_{\text{St}} \in K_2(\mathbb{Z}[G])$ trivial. $\square$

However, Theorem 4.6 has a limitation.

Example 4.7 (Cyclic Groups of Odd Orders) Our formula does not give any information about the K-theory of the integral group ring $\mathbb{Z}[C_n]$ of cyclic group C_n of odd order n . This is because $C_n \otimes C_2 = 0$, so the domain of the map Φ_2 in Equation (4.1.3) is trivial. Because $H_2(BC_n; \mathbb{Z}) = 0$ for all odd numbers n , the short exact sequence in Equation (4.1.1) gives

$$\pi_2(BC_{n+} \wedge \mathbb{K}_Z^{\text{GW}}) \cong K_2(\mathbb{Z}), \quad (4.1.8)$$

and therefore we learn nothing about structures of $K_2(\mathbb{Z}[C_n])$ from this approach, other than it contains $K_2(\mathbb{Z})$ as a subgroup.

4.2. The Steinberg Symbol $\{-1, g\}_{\text{St}}$. We now study the Steinberg symbol $\{-1, g\}_{\text{St}} \in K_2(\mathbb{Z}[G])$, and obtain an expression for it in terms of generators of the Steinberg group $\text{St}(\mathbb{Z}[G])$.

Theorem 4.8

Let G be a group. The Steinberg symbol $\{-1, g\}_{\text{St}} \in K_2(\mathbb{Z}[G])$ for $g \in G$ is given by

$$\{-1, g\}_{\text{St}} = w_{12}(-1) w_{12}(-1) w_{12}(g) w_{12}(g), \quad (4.2.1)$$

where

$$w_{ij}(u) := x_{ij}(u) x_{ji}(-u^{-1}) x_{ij}(u), \quad (4.2.2)$$

and the $x_{ij}(u)$ s are the generators of the Steinberg group $\text{St}(\mathbb{Z}[G])$.

Proof. The proof involves playing with the presentation of the Steinberg group.

We recall from [Ros95, Lemma 4.2.15 on page 195] the identities

$$w_{ij}(u)^{-1} = w_{ij}(-u), \quad (4.2.3)$$

$$w_{ij}(u) = w_{ji}(-u^{-1}), \quad (4.2.4)$$

$$w_{k\ell}(u) w_{ij}(v) w_{k\ell}(u)^{-1} = \begin{cases} w_{ij}(v) & \text{if } i, j, k, \ell \text{ are all distinct,} \\ w_{\ell j}(-u^{-1}v) & \text{if } k = i, \text{ and } i, j, \ell \text{ are all distinct,} \\ w_{i\ell}(-vu) & \text{if } k = j, \text{ and } i, j, k \text{ are all distinct,} \\ w_{ji}(-u^{-1}vu^{-1}) & \text{if } k = i, \text{ and } j = \ell. \end{cases} \quad (4.2.5)$$

In particular, Equation (4.2.5) gives

$$w_{13}(u) w_{12}(v) = w_{23}(v^{-1}u) w_{13}(u), \quad (4.2.6)$$

$$w_{13}(u) w_{23}(v) = w_{12}(-uv^{-1}) w_{13}(u). \quad (4.2.7)$$

Put

$$h_{ij}(u) := w_{ij}(u) w_{ij}(-1), \quad (4.2.8)$$

then a long and boring computation shows

$$\{-1, g\}_{\text{St}} := [h_{12}(-1), h_{13}(g)]$$

$$\begin{aligned}
&= h_{12}(-1) h_{13}(g) h_{12}(-1)^{-1} h_{13}(g)^{-1} \\
&= w_{12}(-1) w_{12}(-1) w_{13}(g) w_{13}(-1) w_{12}(-1)^{-1} w_{12}(-1)^{-1} w_{13}(-1)^{-1} w_{13}(g)^{-1} \\
&= w_{12}(-1) w_{12}(-1) w_{13}(g) w_{13}(-1) w_{12}(1) w_{12}(1) w_{13}(1) w_{13}(-g) \\
&\quad (\text{Equation (4.2.3)}) \\
&= w_{12}(-1) w_{12}(-1) w_{13}(g) \underbrace{w_{13}(-1) w_{12}(1)}_{w_{23}(-1) w_{13}(-1)} w_{12}(1) w_{13}(1) w_{13}(-g) \\
&\quad (\text{Equation (4.2.6)}) \\
&= w_{12}(-1) w_{12}(-1) w_{13}(g) w_{23}(-1) \underbrace{w_{13}(-1) w_{12}(1)}_{w_{23}(-1) w_{13}(-1)} w_{13}(1) w_{13}(-g) \\
&\quad (\text{Equation (4.2.6)}) \\
&= w_{12}(-1) w_{12}(-1) w_{13}(g) w_{23}(-1) w_{23}(-1) \underbrace{w_{13}(-1) w_{13}(1)}_1 w_{13}(-g) \\
&\quad (\text{Equation (4.2.3)}) \\
&= w_{12}(-1) w_{12}(-1) \underbrace{w_{13}(g) w_{23}(-1)}_{w_{12}(g) w_{13}(g)} w_{23}(-1) w_{13}(-g) \\
&\quad (\text{Equation (4.2.7)}) \\
&= w_{12}(-1) w_{12}(-1) w_{12}(g) \underbrace{w_{13}(g) w_{23}(-1) w_{13}(-g)}_{w_{31}(-g^{-1}) w_{23}(-1) w_{31}(g^{-1})} \\
&\quad (\text{Equation (4.2.4)}) \\
&= w_{12}(-1) w_{12}(-1) w_{12}(g) \underbrace{w_{31}(-g^{-1}) w_{23}(-1) w_{31}(g^{-1})}_{w_{21}(-g^{-1})} \\
&\quad (\text{Equation (4.2.5)}) \\
&= w_{12}(-1) w_{12}(-1) w_{12}(g) \underbrace{w_{21}(-g^{-1})}_{w_{12}(g)} \\
&\quad (\text{Equation (4.2.4)}) \\
&= w_{12}(-1) w_{12}(-1) w_{12}(g) w_{12}(g)
\end{aligned}$$

as desired. □

[Theorem 4.8](#) motivates the following definition:

Definition 4.9 (The group $W_{12}(G)$)

Let G be a group. The subgroup $W_{12}(G)$ of $K_2(\mathbb{Z}[G])$ is generated by the Steinberg symbols:

$$W_{12}(G) := \langle \{-1, g\}_{\text{St}} \mid g \in G \rangle. \quad (4.2.9)$$

Notice the following:

- (1) The image of the map

$$\Phi_2 : G \times K_1(\mathbb{Z}) \rightarrow K_2(\mathbb{Z}[G])$$

in [Equation \(4.1.3\)](#) is precisely $W_{12}(G)$.

- (2) Readers who are familiar with the classical definition

$$\text{Wh}_2(G) := \frac{K_2(\mathbb{Z}[G])}{K_2(\mathbb{Z}[G]) \cap W(G)} \quad (4.2.10)$$

of the second Whitehead group proposed by Hatcher-Wagoner in [[HW73](#), page 10] will see our group $W_{12}(G)$ follows their spirit. The group $W(G)$ is generated by the elements $w_{ij}(\pm g)$ for all $g \in G$, and for all i, j , where

$$w_{ij}(u) := x_{ij}(u)x_{ji}(-u^{-1})x_{ij}(u), \quad (4.2.11)$$

and the $x_{ij}(u)$ s' are the free generators of the Steinberg group. See [[Ros05](#), Definition 4.2.1 on page 187, and the end of page 192]. Together with [Theorem 4.8](#), we see that $W_{12}(G)$ is a subgroup of $W(G)$.

- (3) From the functoriality of the Steinberg group, we get a functor

$$W_{12}(-) : \text{Groups} \rightarrow \mathcal{A}\text{belian} \quad (4.2.12)$$

from groups to Abelian groups. Moreover, for every group G , there is a natural group homomorphism

$$\begin{aligned} \Psi_G : G &\rightarrow W_{12}(G) \\ g &\mapsto \{-1, g\}_{\text{St}}, \end{aligned} \quad (4.2.13)$$

which satisfies

$$\Psi_G = \Phi_2|_{\{-1\} \times G} \quad (4.2.14)$$

for Φ_2 as in [Equation \(4.1.3\)](#). Non-injectivity of Ψ_G will imply the non-injectivity of the integral Loday assembly on π_2 .

We are then able to extend the bijectivity result in [Example 4.5](#) to the following injectivity result.

Example 4.10 (Injectivity on $K_2(\mathbb{Z}[C_2^{\oplus n}])$) This example is due to Daniel Ramras.

Denote by C_2 the cyclic group of order 2. Fix $n \in \mathbb{N}$, and consider the n -fold direct sum $C_2^{\oplus n}$. If $x \in C_2^{\oplus n}$ is non-zero, then there exists $1 \leq i \leq n$ such that the image $\text{proj}_i(x)$ of x under the i -th projection map

$$\text{proj}_i : C_2^{\oplus n} \rightarrow C_2$$

is non-zero. Functoriality of $W_{12}(-)$ gives a commutative diagram

$$\begin{array}{ccc} C_2^{\oplus n} & \xrightarrow{\Psi_{C_2^{\oplus n}}} & W_{12}(C_2^{\oplus n}) \\ \text{proj}_i \downarrow & & \downarrow W_{12}(\text{proj}_i) \\ C_2 & \xrightarrow{\Psi_{C_2}} & W_{12}(C_2). \end{array} \quad (4.2.15)$$

We saw in [Example 4.5](#) that Ψ_{C_2} is injective, so the commutativity of [Diagram \(4.2.15\)](#) says $\Psi_{C_2^{\oplus n}}(x) \in W_{12}(C_2^{\oplus n})$ is non-zero.

Now, fix a non-zero element $x \in C_2^{\oplus n}$. There exists $1 \leq i \leq n$ such that $\text{proj}_i(x) \in C_2$ is non-zero. Because Ψ_{C_2} is injective by [Example 4.5](#), we must have

$$\left(W_{12}(\text{proj}_i) \circ \Psi_{C_2^{\oplus n}} \right) (x) \neq 0,$$

and consequently $\Psi_{C_2^{\oplus n}}(x) \neq 0$, proving the group homomorphism

$$\Psi_{C_2^{\oplus n}} : C_2^{\oplus n} \rightarrow W_{12}(C_2^{\oplus n})$$

is injective for every $n \in \mathbb{N}$.

As a result, the Loday assembly map

$$\alpha_{\text{Loday}} : B(C_2^{\oplus n})_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{GW}} \rightarrow \mathbb{K}_{\mathbb{Z}[C_2^{\oplus n}]}^{\text{GW}}$$

for the group $C_2^{\oplus n}$ is injective on π_2 when restricted onto the subgroup $K_2(\mathbb{Z}) \oplus [C_2^{\oplus n} \otimes K_1(\mathbb{Z})]$.

4.3. Kähler Differentials and de Rham Cohomology. Let k be a unital, commutative ring, and A be a unital, commutative k -algebra. We define the **A -module of Kähler differentials** to be the free A -module generated by the symbols da for each $a \in A$, modulo linearity and the Leibniz rule:

$$\Omega_{A|k}^1 := \frac{\langle da \mid a \in A \rangle}{\left\langle \begin{array}{l} d(\lambda a + \mu b) = \lambda da + \mu db, \\ d(ab) = a(db) + b(da) \end{array} \right\rangle}, \quad (4.3.1)$$

where $\lambda, \mu \in k$ and $a, b \in A$. Let 1_k be the multiplicative identity in k . The Leibniz rule implies $d(1_k) = 0$, and consequently, $du = 0$ for any $u \in k$.

We then define the module $\Omega_{A|k}^n$ of **differential n -forms** to be the exterior product

$$\Omega_{A|k}^n := \Lambda_A^n \Omega_{A|k}^1, \quad (4.3.2)$$

where the exterior product is over A , not k . It is spanned by the elements $a_0 da_1 \wedge \cdots \wedge da_n$, for $a_i \in A$, that we usually write $a_0 da_1 \cdots da_n$.

Let us put $\Omega_{A|k}^0 := A$. Then for each $n \geq 0$, the **exterior differential operator** is defined by

$$\begin{aligned} d : \Omega_{A|k}^n &\rightarrow \Omega_{A|k}^{n+1} \\ a_0 da_1 \cdots da_n &\mapsto da_0 da_1 \cdots da_n. \end{aligned} \quad (4.3.3)$$

Since $d(1_k) = 0$, it follows that $d \circ d = 0$, and the following sequence

$$A := \Omega_{A|k}^0 \xrightarrow{d} \Omega_{A|k}^1 \xrightarrow{d} \cdots \xrightarrow{d} \Omega_{A|k}^n \xrightarrow{d} \cdots \quad (4.3.4)$$

is a chain complex, called the **de Rham chain complex of A over k** . The homology groups of the de Rham chain complex are denoted by $H_{\text{dR}}^n(A|k)$, and are the **de Rham cohomology groups of A over k** .

Using the theory of cyclic homology and Chern character, Loday constructed a map

$$K_2(A) \rightarrow H_{\text{dR}}^2(A|k) \quad (4.3.5)$$

that sends the Steinberg symbol $\{x, y\}_{\text{St}}$, where $x, y \in A$, to the cohomology class represented by the differential form $x^{-1}y^{-1}dx dy$ [Lod97, page 275]. One may hope that this map will be useful for studying the Steinberg symbol $\{-1, g\}_{\text{St}}$ when $A = \mathbb{Z}[G]$ is an integral group ring. However, because

$$\begin{aligned} d(-1) &= -d(1) \\ &= 0, \end{aligned}$$

the image of the Steinberg symbol $\{-1, g\}_{\text{St}}$ is always trivial in $H_{\text{dR}}^2(\mathbb{Z}[G]|\mathbb{Z})$. This means we need other methods to analyse $\{-1, g\}_{\text{St}}$.

5. COMPARING ASSEMBLY MAPS

In this section, we show our definition of Loday assembly can be used to compute the Weiss-Williams assembly map (i.e., the universal assembly map).

5.1. Two Intermediate Spectra.

5.1.1. Symmetric Monoidal Categories and the $S^{-1}S$ -Construction.

Definition 5.1 (Symmetric Monoidal Category)

A **symmetric monoidal category** is a category $\mathcal{C}$ together with
(SMC 1) a functor

$$\begin{aligned} \square : \mathcal{C} \times \mathcal{C} &\rightarrow \mathcal{C} \\ (A, B) &\mapsto A \square B, \end{aligned}$$

(SMC 2) a distinguished object $e \in \mathcal{C}$,

(SMC 3) and four basic natural isomorphisms

$$\begin{aligned}
e \square A &\cong A, \\
A \square e &\cong A, \\
(A \square B) \square C &\cong A \square (B \square C), \\
A \square B &\cong B \square A.
\end{aligned}$$

subject to the coherence conditions given in [ML98].

A **monoidal functor** between symmetric monoidal categories is a functor that respects all the axioms. The category of symmetric monoidal categories and monoidal functors will be denoted as SymMonCat .

Example 5.2 The category FinSet of finite sets is a symmetric monoidal category under disjoint union. The distinguished object is the empty set.

Example 5.3 If $(\mathcal{C}, \square)$ is a symmetric monoidal category, then its category $\text{iso}(\mathcal{C})$ of isomorphisms is also a symmetric monoidal category under $\square$.

The axioms (SMC 1)–(SMC 3) make the set $\pi_0(B\mathcal{C})$ of the path components of the classifying space of the symmetric monoidal category $\mathcal{C}$ a monoid. The $S^{-1}S$ -construction is a categorical method to study the group completion of this monoid.

Definition 5.4 (The Category $S^{-1}S$, [Wei13, Definition 4.2 on page 328])

Let $(S, \square)$ be a symmetric monoidal category. We define a new category $S^{-1}S$ from S by the following data:

(SIS 1) Objects in $S^{-1}S$ are pairs (m, n) of objects in S .

(SIS 2) A morphism $(m_1, m_2) \rightarrow (n_1, n_2)$ in $S^{-1}S$ is an equivalence class of composites

$$(m_1, m_2) \xrightarrow{s \square} (s \square m_1, s \square m_2) \xrightarrow{(f, g)} (n_1, n_2).$$

This composite is equivalent to

$$(m_1, m_2) \xrightarrow{t \square} (t \square m_1, t \square m_2) \xrightarrow{(f', g')} (n_1, n_2)$$

if and only if there is an isomorphism $\alpha : s \rightarrow t$ in S so that the composition with $\alpha \square \text{id}_{m_i}$ sends f' and g' to f and g .

We note that a monoidal functor $S \rightarrow T$ induces a functor

$$S^{-1}S \rightarrow T^{-1}T.$$

Moreover, the category $S^{-1}S$ has a natural symmetric monoidal structure induced by the symmetric monoidal structure on S (see [Wei13, Remark 4.2.2 on page 329]). Thus the $S^{-1}S$ -construction gives a functor

$$\text{SymMonCat} \rightarrow \text{SymMonCat}$$

$$S \mapsto S^{-1}S,$$

and $\pi_0(BS^{-1}S)$ is an Abelian group. Under the conditions as in [Wei13, Theorem 4.8 on page 333], $\pi_0(BS^{-1}S)$ is the group completion of $\pi_0(BS)$.

Definition 5.5 (The Category $\mathcal{S}_{\mathcal{A}}$)

Let $\mathcal{A}$ be a symmetric monoidal category. We define the category $\mathcal{S}_{\mathcal{A}}$ to be the $S^{-1}S$ -construction of the category $\text{iso}(\mathcal{A})$ of isomorphisms in $\mathcal{A}$, as defined in Definition 5.4:

$$\mathcal{S}_{\mathcal{A}} := \text{iso}(\mathcal{A})^{-1} \text{iso}(\mathcal{A}) \quad (5.1.1)$$

In particular, we write

$$\mathcal{S}_R := \mathcal{S}_{\text{Free}_R^{\text{fg}}} \quad (5.1.2)$$

for a ring R .

We say a ring R satisfies the **Invariant Basis Property (IBP)** when $R^n \cong R^m$ if and only if $m = n$. The importance of IBP is that we have explicit descriptions for the classifying space $B \text{iso}(\text{Free}_R^{\text{fg}})$ and its homotopy group completion $B\mathcal{S}_R$.

Theorem 5.6 ([Wei13, Theorem 4.9 on page 334])

Let R be a ring satisfying the IBP. Then the classifying space of the category $\text{iso}(\text{Free}_R^{\text{fg}})$ is given by

$$B \text{iso}(\text{Free}_R^{\text{fg}}) = \bigsqcup_{p \geq 1} BGL(p, R). \quad (5.1.3)$$

Moreover, the space $\mathbb{Z} \times BGL(R)^+$ is a model for the classifying space of the $S^{-1}S$ -construction of the category $\text{iso}(\text{Free}_R^{\text{fg}})$:

$$B\mathcal{S}_R \simeq \mathbb{Z} \times BGL(R)^+. \quad (5.1.4)$$

The natural homotopy group completion map

$$\mathbf{gc}_R : \bigsqcup_{p \geq 1} BGL(p, R) \rightarrow \mathbb{Z} \times BGL(R)^+, \quad (5.1.5)$$

induced by the maps sending a matrix $A_p \in GL(p, R)$ to $(p, A_p) \in \mathbb{Z} \times GL(R)$, makes the diagram

$$\begin{array}{ccc} B \text{iso}(\text{Free}_R^{\text{fg}}) & \xrightarrow{\quad} & B\mathcal{S}_R \\ \parallel & & \wr \\ \bigsqcup_{p \geq 1} BGL(p, R) & \xrightarrow{\mathbf{gc}_R} & \mathbb{Z} \times BGL(R)^+ \end{array} \quad (5.1.6)$$

commute up to homotopy. The lightning bolt in the diagram is a zig-zag of homotopy equivalences.

5.1.2. A Categorical Description of the Loday Pairing and the Spectrum $\mathbb{K}_R^{\text{free}}$.

Under [Theorem 5.6](#), we give another formalism of the Loday pairing γ_{Loday} defined in [Definition 2.2](#) using category theory. This formalism is well-documented in [\[Wei81\]](#) and we shall review it below.

Again, let us choose an isomorphism $R^m \otimes S^n \cong (R \otimes S)^{mn}$. Tensor product of modules gives a functor

$$\otimes : \text{iso} \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) \times \text{iso} \left(\mathcal{F}\text{ree}_S^{\text{fg}} \right) \rightarrow \text{iso} \left(\mathcal{F}\text{ree}_{R \otimes S}^{\text{fg}} \right). \quad (5.1.7)$$

We then have the following homotopy commutative diagram

$$\begin{array}{ccc}
 B \text{ iso} \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) \wedge B \text{ iso} \left(\mathcal{F}\text{ree}_S^{\text{fg}} \right) & \xrightarrow{B \otimes} & B \text{ iso} \left(\mathcal{F}\text{ree}_{R \otimes S}^{\text{fg}} \right) \\
 \parallel & & \parallel \\
 \left(\bigsqcup_{p,q \geq 1} BGL(p, R) \times BGL(q, S) \right)_+ & \xrightarrow{B \otimes} & \bigsqcup_{r \geq 0} BGL(r, R \otimes S) \\
 \downarrow & & \downarrow \\
 \left(\bigsqcup_{p,q \geq 1} BGL(p, R)^+ \times BGL(q, S)^+ \right)_+ & \xrightarrow{f} & \bigsqcup_{r \geq 0} BGL(r, R \otimes S)^+ \\
 \parallel & & \downarrow \mathbf{gc}_{R \otimes S}^+ \\
 \left(\bigsqcup_{p \geq 0} BGL(p, R)^+ \right) \wedge \left(\bigsqcup_{q \geq 0} BGL(q, S)^+ \right) & & \\
 \mathbf{gc}_R^+ \wedge \mathbf{gc}_S^+ \downarrow & & \\
 (\mathbb{Z} \times BGL(R)^+) \wedge (\mathbb{Z} \times BGL(S)^+) & \xrightarrow{\gamma_{\text{free}}} & \mathbb{Z} \times BGL(R \otimes S)^+.
 \end{array} \quad (5.1.8)$$

The convention is that $BGL(p, R)^+$ denotes $BGL(p, R)$ for $p \leq 2$; for $p \geq 3$, it denotes the plus construction of $BGL(p, R)$ relative to the subgroup $E(p, R)$ generated by elementary matrices. The map f (resp. the maps $\mathbf{gc}^+$) is induced by $B \otimes$ (resp. $\mathbf{gc}$) via the universal property of the $+$ -construction, which is well-defined up to homotopy.

The group completion $(\mathbb{Z} \times BGL(R)^+, \mathbf{gc}_R)$ satisfies the following **cofinality condition**:

Condition 5.7 (Cofinality Condition)

For every $x := (\xi, \mathcal{M}) \in \mathbb{Z} \times BGL(R)^+$, there exists $m \in \mathbb{N}$ and $x_0 \in BGL(p, R)^+$ such that

$$(\xi, \mathcal{M}) + (m, *_{\infty}) = \mathbf{gc}_R(x_0),$$

where $*_\infty$ is the base-point of $BGL(R)^+$, and the addition operation is the H-group operation of $\mathbb{Z} \times BGL(R)^+$.

We want to define a map

$$\gamma_{\text{free}} : (\mathbb{Z} \times BGL(R)^+) \wedge (\mathbb{Z} \times BGL(S)^+) \rightarrow \mathbb{Z} \times BGL(R \otimes S)^+$$

so that [Diagram \(5.1.8\)](#) commutes up to weak homotopy.

Definition 5.8 (The map $\widetilde{\gamma_{\text{free}}}$)

If $x \in \mathbb{Z} \times BGL(R)^+$ (resp. $y \in \mathbb{Z} \times BGL(S)^+$), let $m \in \mathbb{N}$, $x_0 \in BGL(p, R)^+$ (resp. $n \in \mathbb{N}$, $y_0 \in BGL(q, S)^+$) be as in [Cofinality Condition \(5.7\)](#).

We define a map

$$\widetilde{\gamma_{\text{free}}} : (\mathbb{Z} \times BGL(R)^+) \times (\mathbb{Z} \times BGL(S)^+) \rightarrow \mathbb{Z} \times BGL(R \otimes S)^+$$

by

$$\widetilde{\gamma_{\text{free}}}(x, y) := \mathbf{gc}_{R \otimes S} \circ f(x_0, y_0) - \mathbf{gc}_{R \otimes S} \circ f(*_m, y_0) - \mathbf{gc}_{R \otimes S} \circ f(x_0, *_n) + \mathbf{gc}_{R \otimes S} \circ f(*_m, *_n).$$

Here, $*_m$ (resp. $*_n$) is the base-point of $BGL(m, R)^+$ (resp. $BGL(n, S)^+$). The plus and minus are from the H-group structure of $\mathbb{Z} \times BGL(R \otimes S)^+$.

The choice of isomorphism $R^m \otimes S^n \cong (R \otimes S)^{mn}$ implies the pairing $\widetilde{\gamma_{\text{free}}}$ is well-defined up to weak homotopy.

Lemma 5.9

Under the notations in [Diagram \(5.1.8\)](#), the following diagram

$$\begin{array}{ccc} \left(\bigsqcup_{p \geq 0} BGL(p, R)^+ \right) \times \left(\bigsqcup_{q \geq 0} BGL(q, S)^+ \right) & \xrightarrow{f} & \bigsqcup_{r \geq 0} BGL(r, R \otimes S)^+ \\ \mathbf{gc}_R \wedge \mathbf{gc}_S \downarrow & & \downarrow \mathbf{gc}_{R \otimes S} \\ (\mathbb{Z} \times BGL(R)^+) \times (\mathbb{Z} \times BGL(S)^+) & \xrightarrow{\widetilde{\gamma_{\text{free}}}} & \mathbb{Z} \times BGL(R \otimes S)^+. \end{array}$$

commutes up to homotopy.

Proof. This is just a matter of diagram-chasing. Consider a point x in the image of $\mathbf{gc}_R$. We know

$$x = (p, [M_p])$$

where $[M_p]$ is an equivalence class in $BGL(R)^+$, represented by some $M_p \in BGL(p, R)^+$. The [Cofinality Condition \(5.7\)](#) says there exists $m \in \mathbb{N}$ such that

$$(p, [M_p]) + (m, *_\infty) = (p + m, [M_p])$$

is in the image of $\mathbf{gc}_R$. In fact, we have

$$(p + m, [M_p \boxplus * _m]) = \mathbf{gc}_R(M_p \boxplus * _m),$$

where the operation “ $\boxplus$ ” is induced by block sum of matrices. Similarly, for $y = (q, N_q) \in \mathbb{Z} \times BGL(S)^+$, there exists $n \in \mathbb{N}$ such that

$$(q + n, [N_q \boxplus * _n]) = \mathbf{gc}_S(N_q \boxplus * _n).$$

We then compute

$$\begin{aligned} \widetilde{\gamma_{\text{free}}}(x, y) &= \widetilde{\gamma_{\text{free}}}((p, M_p), (q, N_q)) \\ &:= \mathbf{gc}_{R \otimes S} \circ f(M_p \boxplus * _m, N_q \boxplus * _n) - \mathbf{gc}_{R \otimes S} \circ f(* _m, N_q \boxplus * _n) \\ &\quad - \mathbf{gc}_{R \otimes S} \circ f(M_p \boxplus * _m, * _n) + \mathbf{gc}_{R \otimes S} \circ f(* _m, * _n) \\ &= \mathbf{gc}_{R \otimes S} \circ f(M_p \boxplus * _m, N_q \boxplus * _n) - \mathbf{gc}_{R \otimes S} \circ f(* _m, N_q \boxplus * _n) \\ &\quad - \mathbf{gc}_{R \otimes S} \circ f(M_p \boxplus * _m, * _n) + \mathbf{gc}_{R \otimes S} \circ f(* _m, * _n) \\ &\simeq \mathbf{gc}_{R \otimes S} \circ f(M_p \boxplus * _m, N_q \boxplus * _n) - \mathbf{gc}_{R \otimes S} \circ f(N_q \boxplus * _n, * _m) \\ &\quad - \mathbf{gc}_{R \otimes S} \circ f(M_p \boxplus * _m, * _n) + \mathbf{gc}_{R \otimes S} \circ f(* _m, * _n) \\ &= \mathbf{gc}_{R \otimes S}(M_p \otimes N_q \boxplus M_p^{\boxplus n} \boxplus N_q^{\boxplus m} \boxplus * _{mn}) - \mathbf{gc}_{R \otimes S}(N_q^{\boxplus m} \boxplus * _{mn}) \\ &\quad - \mathbf{gc}_{R \otimes S}(M_p^{\boxplus n} \boxplus * _{mn}) + \mathbf{gc}_{R \otimes S}(* _{mn}) \\ &\simeq \mathbf{gc}_{R \otimes S}(M_p \otimes N_q) \end{aligned}$$

as desired. □

From the definition, it is clear that $\widetilde{\gamma_{\text{free}}}$ is homotopically trivial on the wedge

$$(\mathbb{Z} \times BGL(R)^+) \vee (\mathbb{Z} \times BGL(S)^+),$$

hence $\widetilde{\gamma_{\text{free}}}$ factors through the smash product to give the map

$$\gamma_{\text{free}} : (\mathbb{Z} \times BGL(R)^+) \wedge (\mathbb{Z} \times BGL(S)^+) \rightarrow \mathbb{Z} \times BGL(R \otimes S)^+, \quad (5.1.9)$$

making [Diagram \(5.1.8\)](#) commute up to weak homotopy. In particular, when restricting onto the base-point component $(\{0\} \times BGL(R)^+) \wedge (\{0\} \times BGL(S)^+)$, we have

$$\gamma_{\text{free}}|_{(\{0\} \times BGL(R)^+) \wedge (\{0\} \times BGL(S)^+)} = \gamma_{\text{Loday}} \quad (5.1.10)$$

We are now able to define an intermediate spectrum that allows us to compare the Loday assembly and the universal assembly. Let us mimic the construction given in [Proposition 2.12](#) and recall the map

$$\mathfrak{t}^+ : B \langle t \rangle \rightarrow K_{\Sigma \mathbb{Z}}$$

from in [Definition 2.11](#).

Definition 5.10 (The Spectrum $\mathbb{K}_R^{\text{free}}$)

Let R be a ring satisfying the IBP. Define the spectrum $\mathbb{K}_R^{\text{free}}$ by having

$$\begin{aligned} (\mathbb{K}_R^{\text{free}})_n &:= \mathbb{Z} \times BGL(\Sigma^n R)^+ \\ &\simeq B\mathcal{S}_{\Sigma^n R} \end{aligned} \quad (5.1.11)$$

for $n \geq 0$. Recall our notation that

$$K_R := K_0(R) \times BGL(R)^+.$$

We write

$$K_R^{\text{free}} := \mathbb{Z} \times BGL(R)^+ \quad (5.1.12)$$

The structure maps are given by the composition

$$B\langle t \rangle \wedge K_{\Sigma^n R}^{\text{free}} \xrightarrow{\widehat{\mathfrak{t}}^+ \wedge \text{Id}} K_{\Sigma \mathbb{Z}}^{\text{free}} \wedge K_{\Sigma^n R}^{\text{free}} \xrightarrow{\gamma_{\text{free}}} K_{\Sigma^{n+1} R}^{\text{free}}, \quad (5.1.13)$$

where $\widehat{\mathfrak{t}}^+$ is induced by the group homomorphism

$$\begin{aligned} \langle t \rangle &\rightarrow GL(\Sigma \mathbb{Z}) \\ t &\mapsto \tau, \end{aligned} \quad (5.1.14)$$

and $\tau \in GL(\Sigma \mathbb{Z})$ be the element defined in [Equation \(2.2.4\)](#).

It is worth pointing out that

$$\pi_i(\mathbb{K}_R^{\text{free}}) \cong K_i(R)$$

for all $i \geq 0$. This can be seen from the definition of homotopy groups of a spectrum, or [Theorem 5.14](#) below. However, we do not know if there is a map of spectra

$$\mathbb{K}_R^{\text{free}} \rightarrow \mathbb{K}_R^{\text{GW}}$$

or in the other direction that induces isomorphisms on homotopy groups. It is worth pointing out the following. There is a map of spaces

$$K_R^{\text{free}} \rightarrow K_R \quad (5.1.15)$$

given by

$$\begin{aligned} K_R^{\text{free}} &:= \mathbb{Z} \times BGL(R)^+ \rightarrow K_0(R) \times BGL(R)^+ =: K_R \\ (m, x) &\mapsto ([R^m], x). \end{aligned}$$

This map induces the identity maps on π_n for $n \geq 1$. However, it does not extend to a map of spectra. Otherwise, we would have a commutative diagram

$$\begin{array}{ccccc}
B\langle t \rangle \wedge K_R^{\text{free}} & \xrightarrow{\mathfrak{t}^+ \wedge \text{id}} & K_{\Sigma\mathbb{Z}}^{\text{free}} \wedge K_R^{\text{free}} & \xrightarrow{\gamma_{\text{free}}} & K_{\Sigma R}^{\text{free}} \\
\downarrow & & \downarrow & & \downarrow \\
B\langle t \rangle \wedge K_R & \xrightarrow{\mathfrak{t}^+ \wedge \text{id}} & K_{\Sigma\mathbb{Z}} \wedge K_R & \xrightarrow{\gamma'_{\text{Loday}}} & K_{\Sigma R}
\end{array}$$

of spaces. The right square then induces the following commutative diagram

$$\begin{array}{ccc}
K_i(\Sigma\mathbb{Z}) \otimes K_j(R) & \xrightarrow{\star} & K_{i+j}(\Sigma R) \\
\downarrow & & \downarrow \\
K_i(\Sigma\mathbb{Z}) \otimes K_j(R) & \xrightarrow{\star'} & K_{i+j}(\Sigma R)
\end{array}$$

on homotopy groups. When $i, j \geq 1$, the vertical arrows are identities. The commutativity of this square would imply that

$$[f] \star [g] = [f] \star' [g] \in K_{i+j}(\Sigma R)$$

for all $[f] \in K_i(\Sigma\mathbb{Z})$ and $[g] \in K_j(R)$, which is absurd by [Proposition 2.7](#). As a result, the map in [Equation \(5.1.15\)](#) does not extend to a map of spectra.

Finally, because of the construction of the structure maps in [Equation \(5.1.13\)](#), the pairing γ_{free} constructed in [Equation \(5.1.9\)](#) extends to a weak pairing of spectra. By abuse of notation, we shall denote it as

$$\gamma_{\text{free}} : \mathbb{K}_R^{\text{free}} \wedge \mathbb{K}_S^{\text{free}} \rightarrow \mathbb{K}_{R \otimes S}^{\text{free}} \quad (5.1.16)$$

for any two rings R, S satisfying the IBP.

5.1.3. Idempotent Completion and the Spectrum $\mathbb{K}_R^{\text{proj}}$.

A quick inspection of the classifying space $B\mathcal{S}_R$ given in [Theorem 5.6](#) says

$$\pi_0(B\mathcal{S}_R) \not\cong K_0(R).$$

The spectrum $\mathbb{K}_R^{\text{free}}$ defined in [Definition 5.10](#) fixes this problem at the spectrum-level, and we describe a space-level solution to this problem here.

Definition 5.11 (Idempotent Completion, [Wei13, page 143])

The **idempotent completion** of a category $\mathcal{C}$ is the category $\widehat{\mathcal{C}}$ whose objects are pairs (C, p) with $p : C \rightarrow C$ an idempotent endomorphism of an object C of $\mathcal{C}$.

A morphism $(C, p) \rightarrow (C', p')$ in $\widehat{\mathcal{C}}$ is a map $f : C \rightarrow C'$ in $\mathcal{C}$ such that the diagram

$$\begin{array}{ccc}
C & \xrightarrow{f} & C' \\
p \downarrow & & \downarrow p' \\
C & \xrightarrow{f} & C'
\end{array}$$

(5.1.17)

commutes.

Example 5.12 Because any projective module is a direct summand of a free module, the idempotent completion of $\mathcal{F}\text{ree}_R^{\text{fg}}$ is a (small) category equivalent to the category of finitely generated projective R -modules, and we denote this category by

$$\mathcal{P}\text{roj}_R^{\text{fg}} := \widehat{\mathcal{F}\text{ree}_R^{\text{fg}}} \quad (5.1.18)$$

and call it the category of finitely generated projective R -modules by abuse of terminology.

If $\mathcal{A}$ is a symmetric monoidal category, then so is $\widehat{\mathcal{A}}$. Thus, we define:

Definition 5.13 (The Category $\mathcal{P}_{\mathcal{A}}$)

Let $\mathcal{A}$ be a symmetric monoidal category. We define the category $\mathcal{P}_{\mathcal{A}}$ to be the $S^{-1}S$ -construction of the category $\text{iso}(\widehat{\mathcal{A}})$ of isomorphisms in the idempotent completion $\widehat{\mathcal{A}}$, as defined in [Definition 5.4](#):

$$\mathcal{P}_{\mathcal{A}} := \text{iso}(\widehat{\mathcal{A}})^{-1} \text{iso}(\widehat{\mathcal{A}}) \quad (5.1.19)$$

In other words, $\mathcal{P}_{\mathcal{A}} := \mathcal{S}_{\widehat{\mathcal{A}}}$.

In particular, we write

$$\mathcal{P}_R := \mathcal{P}_{\mathcal{F}\text{ree}_R^{\text{fg}}} \quad (5.1.20)$$

for a ring R .

The following theorem tells us how $\mathcal{S}_R$ and $\mathcal{P}_R$ are related.

Theorem 5.14 ([CP97])

If $\mathcal{C}$ is a category, then there is a canonical embedding

$$\begin{aligned}
\mathcal{C} &\rightarrow \widehat{\mathcal{C}} \\
C &\mapsto (C, \text{id}).
\end{aligned} \quad (5.1.21)$$

Therefore, we have an induced functor

$$\mathcal{S}_{\mathcal{C}} \rightarrow \mathcal{P}_{\mathcal{C}} \quad (5.1.22)$$

between the categories defined in [Definition 5.5](#) and [Definition 5.13](#).

When $\mathcal{C} = \mathcal{F}\text{ree}_R^{\text{fg}}$ is the category of finitely generated free modules over the ring R , the induced group homomorphism

$$\pi_n(B\mathcal{S}_R) \rightarrow \pi_n(B\mathcal{P}_R) \quad (5.1.23)$$

is an isomorphism for $n \geq 1$.

Definition 5.15 (Pairing of Symmetric Monoidal Categories, [May80, page 308])

Let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ be symmetric monoidal categories. A **pairing of symmetric monoidal categories** is a functor

$$\begin{aligned} \otimes : \mathcal{A} \times \mathcal{B} &\rightarrow \mathcal{C} \\ (A, B) &\mapsto A \otimes B \end{aligned} \quad (5.1.24)$$

satisfying the following condition:

(PSMC 1) For any objects $A \in \mathcal{A}$, $B \in \mathcal{B}$, we have

$$0_{\mathcal{A}} \otimes B = 0_{\mathcal{C}} = A \otimes 0_{\mathcal{B}} \quad (5.1.25)$$

(PSMC 2) For any objects $A, A' \in \mathcal{A}$ and $B, B' \in \mathcal{B}$, there is a coherent natural bi-distributivity isomorphism:

$$(A \square_{\mathcal{A}} A') \otimes (B \square_{\mathcal{B}} B') \cong (A \otimes B) \square_{\mathcal{C}} (A \otimes B') \square_{\mathcal{C}} (A' \otimes B) \square_{\mathcal{C}} (A' \otimes B'). \quad (5.1.26)$$

Example 5.16 The standard example of a pairing of symmetric monoidal categories is the tensor product of modules.

Theorem 5.17 ([May80, Theorem 1.6 on page 307, and Theorem 2.1 on page 310])

A pairing $\mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ of symmetric monoidal categories determines a pairing

$$B\mathcal{S}_{\mathcal{A}} \wedge B\mathcal{S}_{\mathcal{B}} \rightarrow B\mathcal{S}_{\mathcal{C}} \quad (5.1.27)$$

of homotopy group completions, which is unique up to homotopy. Moreover, this map fits into the following homotopy commutative diagram:

$$\begin{array}{ccc} B \text{ iso } (\mathcal{A}) \wedge B \text{ iso } (\mathcal{B}) & \longrightarrow & B \text{ iso } (\mathcal{C}) \\ \downarrow & & \downarrow \\ B\mathcal{S}_{\mathcal{A}} \wedge B\mathcal{S}_{\mathcal{A}} & \longrightarrow & B\mathcal{S}_{\mathcal{C}}, \end{array} \quad (5.1.28)$$

where the vertical arrows are the homotopy group completion maps.

Example 5.18 In light of [Theorem 5.6](#) and [Lemma 5.9](#), the map

$$\gamma_{\text{free}} : K_R^{\text{free}} \wedge K_S^{\text{free}} \rightarrow K_{R \otimes S}^{\text{free}}$$

is the pairing of homotopy group completions determined by the tensor product

$$\otimes : \text{iso} \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) \times \text{iso} \left(\mathcal{F}\text{ree}_S^{\text{fg}} \right) \rightarrow \text{iso} \left(\mathcal{F}\text{ree}_{R \otimes S}^{\text{fg}} \right)$$

of modules.

Definition 5.19 (The Pairing γ_{proj})

Let R, S be rings.

We define the pairing

$$\gamma_{\text{proj}} : B\mathcal{P}_R \wedge B\mathcal{P}_S \rightarrow B\mathcal{P}_{R \otimes S} \quad (5.1.29)$$

to be the pairing that is functorially determined by the tensor product

$$\otimes : \text{iso} \left(\mathcal{P}\text{roj}_R^{\text{fg}} \right) \times \text{iso} \left(\mathcal{P}\text{roj}_S^{\text{fg}} \right) \rightarrow \text{iso} \left(\mathcal{P}\text{roj}_{R \otimes S}^{\text{fg}} \right)$$

of modules in the sense of [Theorem 5.17](#).

Because

$$\begin{aligned} B\mathcal{P}_R &\simeq K_0(R) \times BGL(R)^+ \\ &=: K_R \end{aligned} \quad (5.1.30)$$

by [[Wei13](#), Corollary 4.11.1 on page 337], the pairing γ_{proj} induces a map

$$\gamma_{\text{proj}} : K_R \wedge K_S \rightarrow K_{R \otimes S}. \quad (5.1.31)$$

If in addition that the rings R, S both satisfy the IBP, then functoriality of the pairings says the following diagram

$$\begin{array}{ccccc} B \text{ iso} \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) \wedge B \text{ iso} \left(\mathcal{F}\text{ree}_S^{\text{fg}} \right) & \xrightarrow{B \otimes} & B \text{ iso} \left(\mathcal{F}\text{ree}_{R \otimes S}^{\text{fg}} \right) & & \\ \downarrow & \searrow B \text{ (inclusion)} & \downarrow & \searrow B \text{ (inclusion)} & \\ B \text{ iso} \left(\mathcal{P}\text{roj}_R^{\text{fg}} \right) \wedge B \text{ iso} \left(\mathcal{P}\text{roj}_S^{\text{fg}} \right) & \xrightarrow{B \otimes} & B \text{ iso} \left(\mathcal{P}\text{roj}_{R \otimes S}^{\text{fg}} \right) & & \\ \downarrow & & \downarrow & & \\ BS_R \wedge BS_S & \xrightarrow{\gamma_{\text{free}}} & BS_{R \otimes S} & & \\ \downarrow B(5.1.22) & & \downarrow B(5.1.22) & & \\ B\mathcal{P}_R \wedge B\mathcal{P}_S & \xrightarrow{\gamma_{\text{proj}}} & B\mathcal{P}_{R \otimes S} & & \end{array} \quad (5.1.32)$$

commutes up to weak homotopy, where the vertical arrows are the homotopy group completion maps. However, [Proposition 2.7](#) says the induced product maps are different on homotopy groups. Therefore, the pairings γ'_{Loday} and γ_{proj} are not homotopic.

Definition 5.20 (The Spectrum $\mathbb{K}_R^{\text{proj}}$)

Let R be a ring. Define the spectrum $\mathbb{K}_R^{\text{proj}}$ by setting

$$\begin{aligned} \left(\mathbb{K}_R^{\text{proj}} \right)_n &:= K_{\Sigma^n R} \\ &\simeq B\mathcal{P}_{\Sigma^n R} \end{aligned} \tag{5.1.33}$$

for $n \geq 0$. The structure maps are given by the composition

$$B\langle t \rangle \wedge K_R \xrightarrow{\widehat{\mathfrak{t}}^+ \wedge \text{id}} K_{\Sigma \mathbb{Z}} \wedge K_R \xrightarrow{\gamma_{\text{proj}}} K_{\Sigma R}, \tag{5.1.34}$$

where $\widehat{\mathfrak{t}}$ is induced by the group homomorphism in [Equation \(5.1.14\)](#).

Again, the pairing γ_{proj} constructed in [Equation \(5.1.29\)](#) extends to a weak pairing of spectra. By abuse of notation, we shall denote it as

$$\gamma_{\text{proj}} : \mathbb{K}_R^{\text{proj}} \wedge \mathbb{K}_S^{\text{proj}} \rightarrow \mathbb{K}_{R \otimes S}^{\text{proj}} \tag{5.1.35}$$

for any two rings R, S .

The following result is an immediate consequence of [Theorem 5.17](#), [Definition 5.10](#) and [Definition 5.20](#).

Theorem 5.21

Let R, S be rings satisfying the IBP.

The inclusion functor

$$\mathcal{F}\text{ree}_R^{\text{fg}} \rightarrow \mathcal{P}\text{roj}_R^{\text{fg}}$$

induces the following diagram

$$\begin{array}{ccc} \mathbb{K}_R^{\text{free}} \wedge \mathbb{K}_S^{\text{free}} & \xrightarrow{\gamma_{\text{free}}} & \mathbb{K}_{R \otimes S}^{\text{free}} \\ \downarrow & & \downarrow \\ \mathbb{K}_R^{\text{proj}} \wedge \mathbb{K}_S^{\text{proj}} & \xrightarrow{\gamma_{\text{proj}}} & \mathbb{K}_{R \otimes S}^{\text{proj}} \end{array} \tag{5.1.36}$$

of weak pairings of spectra. More precisely, for each pair (m, n) of non-negative integers, the diagram

$$\begin{array}{ccc} K_{\Sigma^m R}^{\text{free}} \wedge K_{\Sigma^n S}^{\text{free}} & \xrightarrow{\gamma_{\text{free}}} & K_{\Sigma^{m+n}(R \otimes S)}^{\text{free}} \\ \downarrow & & \downarrow \\ K_{\Sigma^m R} \wedge K_{\Sigma^n S} & \xrightarrow{\gamma_{\text{proj}}} & K_{\Sigma^{m+n}(R \otimes S)} \end{array} \tag{5.1.37}$$

commutes up to weak homotopy.

The presence of the pairing map γ_{free} allows one to define a multiplication map

$$\begin{aligned} \star_{\text{free}} : K_i(R) \otimes K_j(S) &\rightarrow K_{i+j}(R \otimes S) \\ [f] \otimes [g] &\mapsto [\gamma_{\text{free}} \circ (f \wedge g)] \end{aligned} \quad (5.1.38)$$

It is clear from [Diagram \(5.1.8\)](#) that $\star_{\text{free}}$ recovers tensor product of modules when $i = j = 0$, and coincides with Loday's multiplication $\star$ by [Equation \(5.1.10\)](#) when $i, j \geq 1$. However, the following cases are still open:

Question 5.22

What is the product map

$$\star_{\text{free}} : K_0(R) \otimes K_j(S) \rightarrow K_j(R \otimes S)? \quad (5.1.39)$$

In particular, can we relate this product map with the classical product map

$$\# : K_0(R) \otimes K_1(S) \rightarrow K_1(R \otimes S)$$

defined by Milnor as in [Equation \(2.3.2\)](#) when $j = 1, 2$?

5.2. Categories Over Metric Spaces and the Non-Connective Pedersen-Weibel Algebraic K-Theory Spectrum. Recall the definition of an additive category.

Definition 5.23 (Additive Category)

A category $\mathcal{A}$ is **additive** if all of the following are satisfied:

- (A1) The collection of morphisms $\text{Hom}(A, B)$ from A to B is an Abelian group for each object $A, B \in \mathcal{A}$.
- (A2) The composition of morphisms

$$\text{Hom}(A, B) \times \text{Hom}(B, C) \rightarrow \text{Hom}(A, C)$$

is a bilinear map.

- (A3) There is a distinguished object $0_{\mathcal{A}} \in \mathcal{A}$ such that

$$\text{Hom}(A, 0_{\mathcal{A}}) = 0 = \text{Hom}(0_{\mathcal{A}}, A)$$

for all objects $A \in \mathcal{A}$.

- (A4) There is a binary operation

$$\oplus : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$$

which is both the categorical product and categorical coproduct.

Example 5.24 Our standard example is the category $\mathcal{F}\text{ree}_R^{\text{fg}}$ of finitely generated free modules over the ring R . The binary operation in [\(A4\)](#) is the direct sum of modules. Note that the ring R does not have to satisfy the IBP for $\mathcal{F}\text{ree}_R^{\text{fg}}$ being additive.

Definition 5.25 (Filtered Additive Category, [PW85, Definition 1.1 on page 168])

An additive category $\mathcal{A}$ is said to be **filtered** if there is an increasing filtration

$$F_0(A, B) \subseteq \cdots \subseteq F_n(A, B) \subseteq \cdots \subseteq \text{Hom}(A, B)$$

on $\text{Hom}(A, B)$ for every pair of objects $A, B \in \mathcal{A}$, satisfying the following properties:

(F1) Each $F_i(A, B)$ is a subgroup of $\text{Hom}(A, B)$.

(F2) $\bigcup_i F_i(A, B) = \text{Hom}(A, B)$.

(F3) The identity map id_A and the zero map 0_A are in $F_0(A, A)$.

(F4) The canonical maps

$$A \oplus B \rightarrow A,$$

$$A \rightarrow A \oplus B$$

and all coherence isomorphisms are in $F_0(A, B)$.

(F5) The composition law in $\mathcal{A}$ respects the filtration, meaning that if $f \in F_i(A, B)$ and $g \in F_j(B, C)$, then $g \circ f \in F_{i+j}(A, C)$.

Definition 5.26 (The Category $\mathcal{C}_M(\mathcal{A})$, [Spe04, Definition 4.2 on page 14])

Let $\mathcal{A}$ be a category and (M, d) be a metric space. We definition the category $\mathcal{C}_M(\mathcal{A})$ as follows

- (i) objects are sets $\{A_x\}_{x \in M}$ of objects $A_x \in \mathcal{A}$, with the condition that the collection $\{x \in M \mid A_x \neq 0\}$ is locally finite.
- (ii) A morphism $\varphi : A \rightarrow B$ in $\mathcal{C}_M(\mathcal{A})$ is a collection of morphisms

$$\varphi_y^x : A_x \rightarrow B_y$$

in $\mathcal{A}$ with a bound $R \geq 0$, so that $\varphi_y^x = 0$ whenever $d(x, y) > R$. In this case, we say the morphism $\varphi : A \rightarrow B$ is **bounded** by R .

Note that if $\mathcal{A}$ is an additive category, then so is $\mathcal{C}_M(\mathcal{A})$. Secondly, the boundedness condition on morphisms gives a natural filtration on the Hom-sets of $\mathcal{C}_M(\mathcal{A})$ —for each pair of objects $A, B \in \mathcal{C}_M(\mathcal{A})$, the collection $F_i(A, B)$ consists of morphisms $A \rightarrow B$ in $\mathcal{C}_M(\mathcal{A})$ that are bounded by i . Thus the category $\mathcal{C}_M(\mathcal{A})$ is filtered additive in the sense of Definition 5.25. Thirdly, if the category $\mathcal{A}$ we start with is already filtered, we require all components $\varphi_y^x : A_x \rightarrow B_y$ of the morphism $\varphi : A \rightarrow B$ have filtration degree i as well.

Example 5.27 ($\mathcal{C}_i(\mathcal{A})$) Our standard example for $\mathcal{C}_M(\mathcal{A})$ is the case when $M = \mathbb{Z}^i$. Let us equip $\mathbb{Z}^i$ with the metric induced by the ℓ^∞ -norm, so that

$$\begin{aligned} d(\vec{x}, \vec{y}) &:= \|\vec{x} - \vec{y}\|_{\ell^\infty} \\ &:= \max_{1 \leq j \leq i} |x_j - y_j| \end{aligned}$$

for $\vec{x} = (x_1, \dots, x_i)$, $\vec{y} = (y_1, \dots, y_i) \in \mathbb{Z}^i$. In this case, we write

$$\mathcal{C}_i(\mathcal{A}) := \begin{cases} \mathcal{A} & \text{if } i = 0, \\ \mathcal{C}_{\mathbb{Z}^i}(\mathcal{A}) & \text{if } i > 0 \end{cases} \quad (5.2.1)$$

and we get a sequence $\{\mathcal{C}_i(\mathcal{A})\}_{i=0}^\infty$ of categories. Note that the categories $\mathcal{C}_i(\mathcal{C}_j(\mathcal{A}))$ and $\mathcal{C}_{i+j}(\mathcal{A})$ are isomorphic. The sequence $\{\mathcal{C}_i(\mathcal{A})\}_{i=0}^\infty$ was first constructed by Pedersen-Weibel to produce non-connective spectra from additive categories [PW85]. In particular, when $\mathcal{A}$ is the category of finitely generated free modules, the resulting spectrum has homotopy groups isomorphic to algebraic K-groups of the underlying ring. The categories $\mathcal{C}_M(\mathcal{A})$ were also studied by Carlsson-Pedersen to prove the K-theoretic Novikov Conjecture for a large class of groups [CP95].

Finally, we give $\mathcal{C}_i(\mathcal{A})$ the split exact structure as in [PW85, page 171]. More precisely, We say a chain

$$A \rightarrow C \rightarrow B$$

of morphisms in $\mathcal{C}_1(\mathcal{A})$ is an exact sequence if there is an isomorphism $A \oplus B \xrightarrow{\cong} C$ in $\mathcal{C}_1(\mathcal{A})$ such that the diagram

$$\begin{array}{ccccc} A & \longrightarrow & A \oplus B & \longrightarrow & B \\ \text{id} \downarrow & & \cong \downarrow & & \downarrow \text{id} \\ A & \longrightarrow & C & \longrightarrow & B \end{array} \quad (5.2.2)$$

commutes. Because $\mathcal{C}_i(\mathcal{C}_j(\mathcal{A})) \cong \mathcal{C}_{i+j}(\mathcal{A})$, this gives an exact category structure on $\mathcal{C}_i(\mathcal{A})$ inductively.

Definition 5.28 (The Category $\mathcal{P}_{\mathcal{A},i}$)

Let $\mathcal{A}$ be a split exact, additive category. For each integer $i \geq 0$, we define

$$\mathcal{P}_{\mathcal{A},i} := \mathcal{P}_{\mathcal{C}_i(\mathcal{A})} \quad (5.2.3)$$

as in Definition 5.13. In particular, we write:

$$\mathcal{P}_{R,i} := \mathcal{P}_{\mathcal{C}_i(\mathcal{F}\text{ree}_R^{\text{fg}})} \quad (5.2.4)$$

for a ring R .

Pedersen-Weibel proved the following:

Theorem 5.29 ([PW85, Theorem B on page 167])

Let $\mathcal{A}$ be a split exact, additive category.

For each $i \geq 0$, we define

$$(\mathbb{K}_{\mathcal{A}}^{\text{PW}})_i := B\mathcal{P}_{\mathcal{A},i}. \quad (5.2.5)$$

Then for each $i \geq 0$, we have the delooping

$$(\mathbb{K}_{\mathcal{A}}^{\text{PW}})_i \simeq \Omega \left[(\mathbb{K}_{\mathcal{A}}^{\text{PW}})_{i+1} \right]. \quad (5.2.6)$$

Thus the sequence $\{(\mathbb{K}_{\mathcal{A}}^{\text{PW}})_i\}_{i=0}^{\infty}$ of spaces forms an Ω -spectrum.

Theorem 5.30 (The Non-connective Pedersen-Weibel K-theory Spectrum of Rings, [PW85, Theorem A on page 166])

Let R be a ring.

The spectrum

$$\mathbb{K}_R^{\text{PW}} := \mathbb{K}_{\mathcal{F}\text{ree}_R^{\text{fg}}}^{\text{PW}} \quad (5.2.7)$$

is an Ω -spectrum, the so-called **non-connective Pedersen-Weibel algebraic K-theory spectrum of R** , whose homotopy groups are

$$\pi_i(\mathbb{K}_R^{\text{PW}}) \cong \begin{cases} K_i(R) & \text{if } i \geq 0, \\ \text{negative K-groups of } R \\ \text{defined by Bass in [Bas68]} & \text{if } i < 0. \end{cases} \quad (5.2.8)$$

5.3. The Universal Assembly Map.

Definition 5.31 (Homotopy invariant, excisive, and strongly excisive functors)

We say a functor

$$\mathbb{F} : \mathcal{S}\text{paces} \rightarrow \mathcal{S}\text{pectra}$$

is **homotopy invariant** if it takes homotopy equivalences to homotopy equivalences.

A homotopy invariant functor $\mathbb{F}$ is

- (1) **excisive** if it is excisive and preserves homotopy push-out squares, and $\mathbb{F}(\emptyset)$ is contractible;
- (2) **strongly excisive** if it preserves arbitrary coproducts, up to homotopy equivalence.

Theorem 5.32 (Weiss-Williams/Universal Assembly, [WW95, Theorem 1.1. on page 333])

For any homotopy invariant functor

$$\mathbb{F} : \mathcal{S}\text{paces} \rightarrow \mathcal{S}\text{pectra},$$

there exists a strongly excisive functor

$$\mathbb{F}^{\%} : \mathcal{S}\text{paces} \rightarrow \mathcal{S}\text{pectra}$$

and a natural transformation

$$\alpha_{\text{WW}} : \mathbb{F}^{\%} \Rightarrow \mathbb{F}, \quad (5.3.1)$$

the so-called **Weiss-Williams assembly**, or the **universal assembly**, such that the component

$$\alpha_{\text{WW}} : \mathbb{F}^{\%}(\text{point}) \rightarrow \mathbb{F}(\text{point})$$

of the natural transformation is a homotopy equivalence. Moreover, $\mathbb{F}^\%$ and $\alpha_{\mathbb{W}\mathbb{W}}$ can be made to depend functorially on $\mathbb{F}$.

Proof. (Outline)

The point is that the functor

$$X \mapsto X_+ \wedge \mathbb{F}(\text{point}) \tag{5.3.2}$$

is a model for $\mathbb{F}^\%$; and the natural transformation

$$\alpha_{\mathbb{W}\mathbb{W}} : \mathbb{F}^\% \Rightarrow \mathbb{F}$$

is induced by the constant map $X \rightarrow \text{point}$. □

Theorem 5.33 ([WW95, Observation 1.3 on page 336])

If the homotopy invariant functor

$$\mathbb{F} : \mathbb{S}\text{paces} \rightarrow \mathbb{S}\text{pectra}$$

is already excisive, then the component

$$(\alpha_{\mathbb{W}\mathbb{W}})_X : \mathbb{F}^\%(X) \rightarrow \mathbb{F}(X)$$

of the natural transformation

$$\alpha_{\mathbb{W}\mathbb{W}} : \mathbb{F}^\% \Rightarrow \mathbb{F}$$

is a homotopy equivalence for every compact X .

If $\mathbb{F}$ is strongly excisive, then $(\alpha_{\mathbb{W}\mathbb{W}})_X$ is a homotopy equivalence for all X .

Theorem 5.34 (Universal Property of the Weiss-Williams Assembly, [WW95, page 336])

Suppose the homotopy invariant functor

$$\mathbb{F} : \mathbb{S}\text{paces} \rightarrow \mathbb{S}\text{pectra}$$

admits another natural transformation

$$\beta : \mathbb{E} \Rightarrow \mathbb{F}$$

from a strongly excisive functor $\mathbb{E}$. Then the diagram

$$\begin{array}{ccc} \mathbb{E}^\% & \xrightarrow{\alpha_{\mathbb{W}\mathbb{W}}^\mathbb{E}} & \mathbb{E} \\ \beta^\% \downarrow & & \downarrow \beta \\ \mathbb{F}^\% & \xrightarrow{\alpha_{\mathbb{W}\mathbb{W}}^\mathbb{F}} & \mathbb{F} \end{array} \tag{5.3.3}$$

commutes.

We conclude by the following remark. In Diagram (5.3.3), if the component

$$\beta_{\text{point}} : \mathbb{E}(\text{point}) \rightarrow \mathbb{F}(\text{point})$$

of the natural transformation β is a homotopy equivalence, then

$$\beta_X^{\%} : \mathbb{E}^{\%}(X) \rightarrow \mathbb{F}^{\%}(X)$$

is a homotopy equivalence for all X by the Eilenberg-Steenrod Uniqueness Theorem ([ES52, Theorem 10.1 on page 100–101]). So β is homotopic to $\alpha_{\text{WW}}^{\mathbb{F}}$ after identifying the source and target.

5.4. A Model for the Universal Assembly. We recall the notions of ringoids and modules over them as in [WW95, page 337–338].

Definition 5.35 (Ringoid, [WW95, page 337])

A **ringoid** is a small category in which

- (1) all morphism sets come equipped with an Abelian group structure,
- (2) composition of morphisms is bilinear.

Example 5.36 (The Ringoid $R[\mathcal{C}]$ of the Category $\mathcal{C}$ Over the Ring R) Let R be a ring, and $\mathcal{C}$ be a small category. We define the ringoid $R[\mathcal{C}]$ to be the category having the same objects as $\mathcal{C}$, and the morphism set

$$\text{mor}_{R[\mathcal{C}]}(x, y) := R \langle \text{mor}_{\mathcal{C}}(x, y) \rangle$$

is the free left R -module generated by the set $\text{mor}_{\mathcal{C}}(x, y)$. When $\mathcal{C} = G$ is a group, considered as a category with one object, then the ringoid $R[\mathcal{C}]$ is the group ring $R[G]$, considered as a category with one object, hence justifying the notation.

The category $R[\mathcal{C}]$ is also referred as the **R -category associated to $\mathcal{C}$** in [DL98, page 212].

Definition 5.37 (Modules over Ringoids, [WW95, page 338])

Let $\mathcal{R}$ be a ringoid.

A **left $\mathcal{R}$ -module** is a functor

$$f : \mathcal{R} \rightarrow \text{Abelian}$$

from $\mathcal{R}$ to the category of Abelian groups, such that the induced map

$$f : \text{mor}_{\mathcal{R}}(x, y) \rightarrow \text{mor}_{\text{Abelian}}(f(x), f(y))$$

is a group homomorphism. A **right $\mathcal{R}$ -module** is a left $\mathcal{R}^{op}$ -module.

A left $\mathcal{R}$ -module f is

- (1) **free on one generator** if it is representable:

$$f(-) \cong \text{mor}_{\mathcal{R}}(x, -)$$

for some $x \in \mathcal{R}$;

- (2) **finitely generated free** if it is isomorphic to a finite direct sum of representable ones;
- (3) **finitely generated projective** if it is a direct summand of a finitely generated free one.

The category $\text{Module}_{\mathcal{R}}$ of left $\mathcal{R}$ -modules and natural transformations forms an Abelian category. The subcategory $\text{Free}_{\mathcal{R}}^{\text{fg}}$ of finitely generated free left $\mathcal{R}$ -modules is a split exact, additive category. We can then use the Pedersen-Weibel construction as in [Theorem 5.29](#) to get an Ω -spectrum

$$\mathbb{K}_{\mathcal{R}}^{\text{PW}} := \mathbb{K}_{\text{Free}_{\mathcal{R}}^{\text{fg}}}^{\text{PW}}. \quad (5.4.1)$$

Now, when $\mathcal{R} = R$ is a ring, considered as a category with one object, there is a natural equivalence

$$\text{Free}_{\mathcal{R}}^{\text{fg}} \simeq \text{Free}_R^{\text{fg}}$$

of categories. In this case, the spectra $\mathbb{K}_{\mathcal{R}}^{\text{PW}}$ and $\mathbb{K}_R^{\text{PW}}$ are weakly equivalent.

Let us construct a functor $\mathbb{Y} : \text{Spaces} \rightarrow \text{Spectra}$. Let $\Pi(X)$ be the fundamental groupoid of the topological space X . Given a ring R , we can form the ringoid $R[\Pi(X)]$ as in [Example 5.36](#). Our functor is then given by:

Definition 5.38 (The Functor $\mathbb{Y}$)

Let R be a ring. We define the functor $\mathbb{Y}$ by

$$\begin{aligned} \mathbb{Y} : \text{Spaces} &\rightarrow \text{Spectra} \\ X &\mapsto \mathbb{K}_{R[\Pi(X)]}^{\text{PW}} \end{aligned} \quad (5.4.2)$$

Clearly, the functor $\mathbb{Y}$ is homotopy invariant, so [Theorem 5.33](#) asserts the existence of the universal assembly map

$$\alpha_{\text{WW}} : \mathbb{Y}^{\%} \rightarrow \mathbb{Y}.$$

We will construct this assembly map explicitly following [\[Spe04\]](#). The key ingredient is an analogue of the Loday pairing for the Pedersen-Weibel spectrum.

Lemma 5.39 ([Spe04, Lemma 4.6 on page 16 and Lemma 4.7 on page 17])

Let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{C}$ be split exact, additive categories, considered as symmetric monoidal categories in the canonical way.

Suppose $\otimes : \mathcal{A} \times \mathcal{B} \rightarrow \mathcal{C}$ is a pairing of symmetric monoidal categories. Then

- (i) for all $i, j \geq 0$, there is an induced pairing

$$\otimes_j^i : \mathcal{C}_i(\mathcal{A}) \times \mathcal{C}_j(\mathcal{B}) \rightarrow \mathcal{C}_{i+j}(\mathcal{C}) \quad (5.4.3)$$

of symmetric monoidal categories.

- (ii) Moreover, the collection $\{\otimes\}_{i,j=0}^{\infty}$ of pairings assemble to give a pairing

$$\widehat{\otimes} : \mathbb{K}_{\mathcal{A}}^{\text{PW}} \wedge \mathbb{K}_{\mathcal{B}}^{\text{PW}} \rightarrow \mathbb{K}_{\mathcal{C}}^{\text{PW}}$$

of spectra.

Let us apply this result to ringoids.

Definition 5.40 (Tensor Product of Ringoids)

Let $\mathcal{R}, \mathcal{S}$ be ringoids. The tensor product of $\mathcal{R}$ and $\mathcal{S}$ is the ringoid $\mathcal{R} \otimes \mathcal{S}$ whose objects are given by pairs (r, s) of objects in $\mathcal{R}$ and $\mathcal{S}$, and the collection of morphisms is given by

$$\mathrm{Hom}_{\mathcal{R} \otimes \mathcal{S}}((r_1, s_1), (r_2, s_2)) := \mathrm{Hom}_{\mathcal{R}}(r_1, r_2) \otimes_{\mathbb{Z}} \mathrm{Hom}_{\mathcal{S}}(s_1, s_2). \quad (5.4.4)$$

Composition is given by

$$(f_1 \otimes f_2) \circ (g_1 \otimes g_2) := (f_1 \circ g_1) \otimes (f_2 \circ g_2), \quad (5.4.5)$$

and then extended linearly.

When the ringoids are rings considered as categories with one object, this tensor product becomes tensor product of rings in the usual sense. Furthermore, the tensor product of ringoids extends to a tensor product of modules over ringoids as defined in [Definition 5.37](#).

Example 5.41 Let R be a ring, and $\mathcal{C}$ be a small category. We can then form the ringoids $\mathbb{Z}[\mathcal{C}]$, $R[\mathcal{C}]$ as in [Example 5.36](#). Tensor product of ringoids then gives

$$R \otimes \mathbb{Z}[\mathcal{C}] \cong R[\mathcal{C}]. \quad (5.4.6)$$

Moreover, it also gives a pairing

$$\otimes : \mathrm{Free}_R^{\mathrm{fg}} \times \mathrm{Free}_{\mathbb{Z}[\mathcal{C}]}^{\mathrm{fg}} \rightarrow \mathrm{Free}_{R[\mathcal{C}]}^{\mathrm{fg}} \quad (5.4.7)$$

of symmetric monoidal categories as in [Definition 5.15](#). Note that in this case, we are considering modules over ringoids as in [Definition 5.37](#).

We are now ready to construct a model for the universal assembly map associated to the functor $\mathbb{Y}$. Let us recall from the proof of [Theorem 5.32](#) that the functor

$$X \mapsto X_+ \wedge \mathbb{Y}(\mathrm{point})$$

is a model for the functor $\mathbb{Y}^{\%} : \mathcal{S}\mathrm{paces} \rightarrow \mathcal{S}\mathrm{pectra}$.

Definition 5.42 (The Assembly $\hat{\alpha}$)

Write $\Pi(X)$ to be the fundamental groupoid for the topological space X . Let R be a ring.

Define a map of spectra

$$\hat{\alpha}_X : X_+ \wedge \mathbb{K}_R^{\mathrm{PW}} \rightarrow \mathbb{K}_{R[\Pi(X)]}^{\mathrm{PW}}$$

having components

$$\begin{array}{ccc}
X_+ \wedge (\mathbb{K}_R^{\text{PW}})_n & \xrightarrow{c_+ \wedge \text{id}} B\Pi(X)_+ \wedge (\mathbb{K}_R^{\text{PW}})_n & \xrightarrow{J \wedge \text{id}} (\mathbb{K}_{\mathbb{Z}[\Pi(X)]}^{\text{PW}})_{0+} \wedge (\mathbb{K}_R^{\text{PW}})_n \\
& \searrow (\hat{\alpha}_X)_n & \downarrow \hat{\otimes} \\
& & (\mathbb{K}_{R[\Pi(X)]}^{\text{PW}})_n.
\end{array} \tag{5.4.8}$$

Here,

- (i) the map $c : X \rightarrow B\Pi(X)$ is the classifying map of the universal cover of X ;
- (ii) the map $J : B\Pi(X)_+ \rightarrow (\mathbb{K}_{\mathbb{Z}[\Pi(X)]}^{\text{PW}})_{0+}$ is induced by the inclusion functor

$$\Pi(X) \rightarrow \text{iso}(\mathbb{Z}[\Pi(X)]);$$

- (iii) and the map $\hat{\otimes}$ is the pairing map constructed in [Lemma 5.39](#). In particular, it is induced by the tensor product of ringoids as constructed in [Definition 5.40](#).

The collection of maps $\{\hat{\alpha}_X \mid X \in \text{Spaces}\}$ then gives a natural transformation

$$\hat{\alpha} : \mathbb{Y}^\% \Rightarrow \mathbb{Y} \tag{5.4.9}$$

of the functor $\mathbb{Y} : \text{Spaces} \rightarrow \text{Spectra}$ constructed in [Definition 5.38](#).

Theorem 5.43 ([\[HP04, Theorem 4.3 on page 39\]](#), [\[Spe04, Theorem 4.9 on page 18\]](#))

The natural transformation $\hat{\alpha} : \mathbb{Y}^\% \Rightarrow \mathbb{Y}$ in [Definition 5.42](#) is (homotopic to) the universal assembly map in the sense of [Theorem 5.32](#).

5.5. The Comparison. We mimic [Definition 3.1](#) to create two Loday-like assemblies.

Definition 5.44 (The Assembly Maps α_{free} and α_{proj})

Let R be a ring and G be a group.

We define two maps of spectra

$$BG \wedge \mathbb{K}_R^{\text{free}} \rightarrow \mathbb{K}_{R[G]}^{\text{free}} \tag{5.5.1}$$

$$BG \wedge \mathbb{K}_R^{\text{proj}} \rightarrow \mathbb{K}_{R[G]}^{\text{proj}} \tag{5.5.2}$$

having components

$$\begin{array}{ccc}
BG \wedge (\mathbb{K}_R^{\text{free}})_n & \xrightarrow{j^{\text{free}} \wedge \text{id}} (\mathbb{K}_{\mathbb{Z}[G]}^{\text{free}})_0 \wedge (\mathbb{K}_R^{\text{free}})_n & BG \wedge (\mathbb{K}_R^{\text{proj}})_n \xrightarrow{j^+ \wedge \text{id}} (\mathbb{K}_{\mathbb{Z}[G]}^{\text{proj}})_0 \wedge (\mathbb{K}_R^{\text{proj}})_n \\
& \searrow (\alpha_{\text{free}})_n & \downarrow \gamma_{\text{free}} \\
& & (\mathbb{K}_{R[G]}^{\text{free}})_n, \\
& & \downarrow \gamma_{\text{proj}} \\
& & (\mathbb{K}_{R[G]}^{\text{proj}})_n
\end{array} \tag{5.5.3}$$

respectively. Here, the map j^+ is defined in [Equation \(3.1.3\)](#), and the map j^{free} is defined analogously. We then use the construction in [Definition 3.3](#) to extend these two maps and get:

$$\alpha_{\text{free}} : BG_+ \wedge \mathbb{K}_R^{\text{free}} \rightarrow \mathbb{K}_{R[G]}^{\text{free}}, \quad (5.5.4)$$

$$\alpha_{\text{proj}} : BG_+ \wedge \mathbb{K}_R^{\text{proj}} \rightarrow \mathbb{K}_{R[G]}^{\text{proj}}. \quad (5.5.5)$$

In light of their definitions and [Proposition 3.2](#), the maps α_{free} and α_{proj} are maps of spectra, well-defined up to weak-homotopy. The following result is an immediate consequence of the definitions.

Proposition 5.45

Let R be a ring satisfying the IBP, and let G be a group.

The canonical map

$$\mathcal{S}_R \rightarrow \mathcal{P}_R$$

given in [Equation \(5.1.22\)](#) gives a weak equivalence

$$\mu : \mathbb{K}_R^{\text{free}} \rightarrow \mathbb{K}_R^{\text{proj}} \quad (5.5.6)$$

of spectra. Moreover, this weak equivalence fits into the following diagram

$$\begin{array}{ccc} BG_+ \wedge \mathbb{K}_R^{\text{free}} & \xrightarrow{\alpha_{\text{free}}} & \mathbb{K}_{R[G]}^{\text{free}} \\ \mu \downarrow & & \downarrow \mu \\ BG_+ \wedge \mathbb{K}_R^{\text{proj}} & \xrightarrow{\alpha_{\text{proj}}} & \mathbb{K}_{R[G]}^{\text{proj}} \end{array} \quad (5.5.7)$$

of spectra, which commutes up to weak homotopy.

The proof of the following result was outlined in [\[HP04\]](#), and later completed in [\[Spe04\]](#).

Theorem 5.46

Let R be a ring satisfying the IBP.

For each $i \geq 0$, there is a functor

$$G_i : \mathcal{C}_i \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) \rightarrow \mathcal{F}\text{ree}_{\Sigma^i R}^{\text{fg}}, \quad (5.5.8)$$

such that

- (a) the collection $\{G_i\}_{i=0}^{\infty}$ of functors gives a homotopy equivalence

$$\mathfrak{g} : \mathbb{K}_R^{\text{PW}} \rightarrow \mathbb{K}_R^{\text{proj}} \quad (5.5.9)$$

of spectra [\[Spe04, Theorem 5.14 on page 26 and Proposition 5.15 on page 28\]](#).

- (b) Moreover, the map $\mathfrak{g}$ is compatible with the pairings of spectra. More precisely, for each pair of non-negative integers m, n , there is a diagram

$$\begin{array}{ccc}
(\mathbb{K}_R^{\text{PW}})_m \wedge (\mathbb{K}_R^{\text{PW}})_n & \xrightarrow{\widehat{\otimes}} & (\mathbb{K}_R^{\text{PW}})_{m+n} \\
\mathfrak{g}_m \wedge \mathfrak{g}_n \downarrow & & \downarrow \mathfrak{g}_{m+n} \\
(\mathbb{K}_R^{\text{proj}})_m \wedge (\mathbb{K}_R^{\text{proj}})_n & \xrightarrow{\gamma_{\text{proj}}} & (\mathbb{K}_R^{\text{proj}})_{m+n}
\end{array} \tag{5.5.10}$$

that commutes up to weak homotopy [Spe04, Corollary 6.3 on page 40].

Proof. (Outline) We outline the construction of the functors G_n here since this theorem is the main result of [Spe04]. We begin by constructing the functors

$$G_n : \mathcal{C}_n \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) \rightarrow \mathcal{F}\text{ree}_{\Sigma^n R}^{\text{fg}}$$

inductively.

Step 1: When $n = 0$, we have

$$\mathcal{C}_0 \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) = \mathcal{F}\text{ree}_R^{\text{fg}}.$$

Thus, we define

$$G_0 := \text{id}_{\mathcal{F}\text{ree}_R^{\text{fg}}}. \tag{5.5.11}$$

Step 2: Secondly, let $\mathcal{F}\text{ree}_R^{\mathbb{N}}$ be the category of countably generated free R -modules and locally finite matrices over R . It is proven that the categories

$$\frac{\mathcal{F}\text{ree}_R^{\mathbb{N}}}{\mathcal{F}\text{ree}_R^{\text{fg}}} \simeq \mathcal{F}\text{ree}_{\Sigma R}^{\text{fg}}$$

are equivalent [PW89, Proposition 6.1 on page 359].

The idea is as follows: choose an infinitely based R -module R^∞ in $\mathcal{F}\text{ree}_R^{\mathbb{N}}$, and observe that

$$\text{End}_{\mathcal{F}\text{ree}_R^{\mathbb{N}}} (R^\infty) \cong \text{cone}(R).$$

Now, the category $\mathcal{F}\text{ree}_R^{\mathbb{N}}$ is Karoubi-filtered by its subcategory $\mathcal{F}\text{ree}_R^{\text{fg}}$, and the completely continuous endomorphisms of R^∞ form the ideal $m(R)$ of $\text{cone}(R)$. Thus, we have

$$\text{End}_{\frac{\mathcal{F}\text{ree}_R^{\mathbb{N}}}{\mathcal{F}\text{ree}_R^{\text{fg}}}} (R^\infty) \cong \Sigma R.$$

The canonical additive functor

$$\begin{aligned}
\mathcal{F}\text{ree}_{\Sigma R}^{\text{fg}} &\rightarrow \frac{\mathcal{F}\text{ree}_R^{\mathbb{N}}}{\mathcal{F}\text{ree}_R^{\text{fg}}} \\
\Sigma R &\mapsto R^\infty
\end{aligned} \tag{5.5.12}$$

is therefore full and faithful. But every object of $\frac{\mathcal{F}\text{ree}_R^{\mathbb{N}}}{\mathcal{F}\text{ree}_R^{\text{fg}}}$ is either isomorphic to the zero module or to R^∞ , so this functor is also an equivalence.

We denote by

$$\alpha : \frac{\mathcal{F}\text{ree}_R^{\mathbb{N}}}{\mathcal{F}\text{ree}_R^{\text{fg}}} \rightarrow \mathcal{F}\text{ree}_{\Sigma R}^{\text{fg}} \quad (5.5.13)$$

the inverse equivalence functor of Equation (5.5.12).

Step 3: Thirdly, we define the functor

$$\begin{aligned} \beta : \mathcal{C}_+ \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) &\rightarrow \mathcal{F}\text{ree}_R^{\mathbb{N}} \\ \{A_i\}_{i=0}^\infty &\mapsto \bigoplus_{i=0}^\infty A_i, \end{aligned}$$

for which $\mathcal{C}_+ \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right)$ is the subcategory of $\mathcal{C}_1 \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right)$ consisting objects $\{A_i\}_{i=-\infty}^\infty$ with $A_i = 0$ whenever $i < 0$.

Note that each A_i is finitely generated, so the direct sum $\bigoplus_{i=0}^\infty A_i$ is an object in $\mathcal{F}\text{ree}_R^{\mathbb{N}}$. Moreover, this functor induces a functor

$$\widehat{\beta} : \frac{\mathcal{C}_+ \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right)}{\mathcal{F}\text{ree}_R^{\text{fg}}} \rightarrow \frac{\mathcal{F}\text{ree}_R^{\mathbb{N}}}{\mathcal{F}\text{ree}_R^{\text{fg}}}. \quad (5.5.14)$$

We comment that $\mathcal{F}\text{ree}_R^{\text{fg}}$ is regarded as a subcategory of $\mathcal{C}_+ \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right)$ via the embedding

$$A \mapsto (A, 0, 0, \dots).$$

Step 4: Next, for a filtered additive category $\mathcal{A}$ in the sense of Definition 5.25, we define the functor

$$\begin{aligned} \tau : \mathcal{C}_1(\mathcal{A}) &\rightarrow \frac{\mathcal{C}_+(\mathcal{A})}{\mathcal{A}} \\ \{A_i\}_{i=-\infty}^\infty &\mapsto \{A_i\}_{i=0}^\infty. \end{aligned} \quad (5.5.15)$$

Recall that a morphism ϕ in $\mathcal{C}_1(\mathcal{A})$ is an infinite matrix $(\phi_{i,j})_{i,j=-\infty}^\infty$. We define $\tau(\phi)$ to be the sub-matrix $(\phi_{i,j})_{i,j=1}^\infty$.

Step 5: We now define the functor

$$G_n : \mathcal{C}_n \left(\mathcal{F}\text{ree}_R^{\text{fg}} \right) \rightarrow \mathcal{F}\text{ree}_{\Sigma^n R}^{\text{fg}}$$

for $n \geq 1$ inductively. When $n = 1$, we define G_1 to be the composition

$$\begin{array}{ccccccc}
\mathcal{C}_1(\mathcal{F}\text{ree}_R^{\text{fg}}) & \xrightarrow{\tau} & \frac{\mathcal{C}_+(\mathcal{F}\text{ree}_R^{\text{fg}})}{\mathcal{F}\text{ree}_R^{\text{fg}}} & \xrightarrow{\widehat{\beta}} & \frac{\mathcal{F}\text{ree}_R^{\text{N}}}{\mathcal{F}\text{ree}_R^{\text{fg}}} & \xrightarrow{\alpha} & \mathcal{F}\text{ree}_{\Sigma R}^{\text{fg}} \\
& & & & & \nearrow & \\
& & & & & G_1 &
\end{array} \tag{5.5.16}$$

Now suppose the functor

$$G_n : \mathcal{C}_n(\mathcal{F}\text{ree}_R^{\text{fg}}) \rightarrow \mathcal{F}\text{ree}_{\Sigma^n R}^{\text{fg}}$$

is defined. As pointed out in [Example 5.27](#), the categories $\mathcal{C}_i(\mathcal{C}_j(\mathcal{F}\text{ree}_R^{\text{fg}}))$ and $\mathcal{C}_{i+j}(\mathcal{F}\text{ree}_R^{\text{fg}})$ are isomorphic. We define the functor G_{n+1} to be the composition:

$$\begin{array}{ccc}
\mathcal{C}_{n+1}(\mathcal{F}\text{ree}_R^{\text{fg}}) & \xrightarrow{\cong} & \mathcal{C}_1(\mathcal{C}_n(\mathcal{F}\text{ree}_R^{\text{fg}})) \xrightarrow{\mathcal{C}_1(G_n)} \mathcal{C}_1(\mathcal{F}\text{ree}_{\Sigma^n R}^{\text{fg}}) \\
& & \downarrow \tau \\
& & \frac{\mathcal{C}_+(\mathcal{F}\text{ree}_{\Sigma^n R}^{\text{fg}})}{\mathcal{F}\text{ree}_{\Sigma^n R}^{\text{fg}}} \\
& & \downarrow \widehat{\beta} \\
& & \frac{\mathcal{F}\text{ree}_{\Sigma^n R}^{\text{N}}}{\mathcal{F}\text{ree}_{\Sigma^n R}^{\text{fg}}} \\
& & \downarrow \alpha \\
& & \mathcal{F}\text{ree}_{\Sigma^{n+1} R}^{\text{fg}} \\
& \searrow G_{n+1} & \\
& &
\end{array} \tag{5.5.17}$$

This completes the construction of the functors

$$G_i : \mathcal{C}_i(\mathcal{F}\text{ree}_R^{\text{fg}}) \rightarrow \mathcal{F}\text{ree}_{\Sigma^i R}^{\text{fg}},$$

and we refer readers to [\[Spe04, Theorem 5.14 on page 26, Proposition 5.15 on page 28 and Corollary 6.3 on page 40\]](#) for checking the remaining properties. $\square$

A consequence of [Theorem 5.46](#) is the following corollary:

Corollary 5.47

Let R be a ring satisfying IBP, and G be a group.

Consider G as a groupoid in the canonical way. Then the component

$$\widehat{\alpha}_{BG} : BG_+ \wedge \mathbb{K}_R^{\text{PW}} \rightarrow \mathbb{K}_{R[G]}^{\text{PW}} \tag{5.5.18}$$

of the universal assembly map $\hat{\alpha}$, as constructed in [Definition 5.42](#), is weakly homotopic to the assembly map

$$\alpha_{\text{proj}} : BG_+ \wedge \mathbb{K}_R^{\text{proj}} \rightarrow \mathbb{K}_{R[G]}^{\text{proj}} \quad (5.5.19)$$

constructed in [Definition 5.44](#).

Proof. We need to check the diagram

$$\begin{array}{ccccccc} BG_+ \wedge (\mathbb{K}_R^{\text{PW}})_n & \xrightarrow{c_+ \wedge \text{id}} & BG_+ \wedge (\mathbb{K}_R^{\text{PW}})_n & \xrightarrow{J \wedge \text{id}} & (\mathbb{K}_{\mathbb{Z}[G]}^{\text{PW}})_0 \wedge (\mathbb{K}_R^{\text{PW}})_n & \xrightarrow{\hat{\otimes}} & (\mathbb{K}_{R[G]}^{\text{PW}})_n \\ \text{id} \wedge \mathfrak{g}_n \downarrow & & \text{id} \wedge \mathfrak{g}_n \downarrow & & \text{id} \wedge \mathfrak{g}_n \downarrow & & \downarrow \mathfrak{g}_n \\ BG_+ \wedge (\mathbb{K}_R^{\text{proj}})_n & \xrightarrow{c_+ \wedge \text{id}} & BG_+ \wedge (\mathbb{K}_R^{\text{proj}})_n & \xrightarrow{j^+ \wedge \text{id}} & (\mathbb{K}_{\mathbb{Z}[G]}^{\text{proj}})_0 \wedge (\mathbb{K}_R^{\text{proj}})_n & \xrightarrow{\gamma_{\text{proj}}} & (\mathbb{K}_{R[G]}^{\text{proj}})_n \end{array} \quad (5.5.20)$$

commutes up to weak homotopy for all $n \geq 0$. Here,

(1) the classifying maps

$$c_+ : BG_+ \rightarrow BG_+$$

for the universal cover of BG_+ is just the identity, so the left square commutes up to homotopy automatically.

(2) The middle square commutes up to homotopy by the definitions of J and j^+ .

(3) The right square commutes up to weak homotopy by part (b) of [Theorem 5.46](#).

Finally, the map $\mathfrak{g}_n$ is a homotopy equivalence by [Theorem 5.46 \(a\)](#). So the claim holds as desired. $\square$

Corollary 5.48 (Corollary of [Corollary 5.47](#))

Let R be a ring satisfying IBP, and G be a group.

Consider G as a groupoid in the canonical way, then the component

$$\hat{\alpha}_{BG} : BG_+ \wedge \mathbb{K}_R^{\text{PW}} \rightarrow \mathbb{K}_{R[G]}^{\text{PW}} \quad (5.5.21)$$

of the universal assembly map $\hat{\alpha}$, as constructed in [Definition 5.42](#), is weakly homotopic to the assembly map

$$\alpha_{\text{free}} : BG_+ \wedge \mathbb{K}_R^{\text{free}} \rightarrow \mathbb{K}_{R[G]}^{\text{free}} \quad (5.5.22)$$

constructed in [Definition 5.44](#).

5.6. An Explicit Formula for the Universal Assembly. We now relate [Theorem 3.8](#) to the universal assembly map.

Theorem 5.49

Let R be a ring and G be a group.

There is a subgroup $\text{AHSS}(BG)_{1,i}^\infty$ of $\pi_{i+1}(BG_+ \wedge \mathbb{K}_R^{\text{PW}})$ which is a quotient of the E^2 -term

$$\mathrm{AHSS}(BG)_{1,i}^2 \cong G_{ab} \otimes K_i(R)$$

from the Atiyah-Hirzebruch spectral sequence of the classifying space BG of G with coefficients in the Pedersen-Weibel K-theory spectrum $\mathbb{K}_R^{\mathrm{PW}}$ of R .

For $i > 0$, the filler of the diagram

$$\begin{array}{ccc} \mathrm{AHSS}(BG)_{1,i}^2 \cong G_{ab} \otimes K_i(R) & \xrightarrow{\widehat{\Phi}_i} & K_{i+1}(R[G]) \\ \downarrow & & \uparrow \pi_{i+1}(\widehat{\alpha}_{BG}) \\ \mathrm{AHSS}(BG)_{1,i}^\infty & \hookrightarrow & \pi_{i+1}(BG \wedge \mathbb{K}_R^{\mathrm{PW}}) \end{array} \quad (5.6.1)$$

is induced by

$$\begin{aligned} G \times K_i(R) &\rightarrow K_{i+1}(R[G]) \\ (g, [f]) &\mapsto [f] \star \{g\}. \end{aligned} \quad (5.6.2)$$

This provides an explicit formula for the universal assembly map when restricted onto the subgroup $\mathrm{AHSS}(BG)_{1,i}^\infty$.

Proof. From [Corollary 5.48](#), we know the universal assembly $\widehat{\alpha}_{BG}$ induces the same map on homotopy groups as the assembly map α_{free} . So we need to verify the formula for α_{free} .

As pointed out in [Equation \(5.1.10\)](#), the pairing map γ_{free} used in constructing the assembly α_{free} is homotopic to the original Loday pairing γ_{Loday} . Therefore, by repeating the same proofs as in [Proposition 3.6](#) and [Proposition 3.7](#), we see that the bottom composition of

$$\begin{array}{ccccc} \pi_{i+1} \left(\left(\bigvee_{g \in G} \mathbb{S}^1 \right) \wedge \mathbb{K}_R^{\mathrm{free}} \right) & \xrightarrow{(i_G)_*} & \pi_{i+1}(BG \wedge \mathbb{K}_R^{\mathrm{free}}) & \xrightarrow{\alpha_{\mathrm{free}}} & K_{i+1}(R[G]) \\ \parallel & & \parallel & & \parallel \\ \bigoplus_{g \in G} K_i(R) & \longrightarrow & \pi_{i+1}(BG \wedge \mathbb{K}_R^{\mathrm{free}}) & \longrightarrow & K_{i+1}(R[G]) \end{array} \quad (5.6.3)$$

sends the element $[f] \in K_i(R)$ in the summand in $\bigoplus_{g \in G} K_i(R)$ labelled by $g \in G$ to the element

$$[f] \star \{g\} \in K_{i+1}(R[G]).$$

This completes the proof. $\square$

The universal assembly then admits the following version of [Corollary 4.4](#).

Corollary 5.50 (The Universal Assembly on π_2 of an Integral Group Ring)

Let G be a group. The universal assembly map

$$\widehat{\alpha}_{BG} : BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\mathrm{PW}} \rightarrow \mathbb{K}_{\mathbb{Z}[G]}^{\mathrm{PW}} \quad (5.6.4)$$

for the integral group ring $\mathbb{Z}[G]$ on π_2 , when restricted onto the subgroup

$$K_2(\mathbb{Z}) \oplus [G_{ab} \oplus K_1(\mathbb{Z})],$$

is given by the formula:

$$K_2(i) \oplus \widehat{\Phi}_2 : K_2(\mathbb{Z}) \oplus [G_{ab} \oplus K_1(\mathbb{Z})] \rightarrow K_2(\mathbb{Z}[G]), \quad (5.6.5)$$

for which $i : \mathbb{Z} \rightarrow \mathbb{Z}[G]$ is the inclusion, and $\widehat{\Phi}_2$ is induced by the map

$$\begin{aligned} G \times K_1(\mathbb{Z}) &\rightarrow K_2(\mathbb{Z}[G]) \\ (g, \pm 1) &\mapsto -\{\pm 1, g\}_{\text{st}}. \end{aligned} \quad (5.6.6)$$

Proof. We have $-\{\pm 1, g\}_{\text{st}}$ instead of $\{\pm 1, g\}_{\text{st}}$ because of [Proposition 2.4](#) (iii). $\square$

Because of [Question 5.22](#), it is unclear if [Corollary 3.9](#) (or its variants) holds for the universal assembly. However, Waldhausen showed that when $R = \mathbb{Z}$, the assembly

$$\pi_1(\alpha_{\text{free}}) : \pi_1(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{free}}) \rightarrow K_1(\mathbb{Z}[G])$$

on π_1 is the usual map

$$\begin{aligned} \{1, -1\} \oplus G_{ab} &\rightarrow K_1(\mathbb{Z}[G]) \\ (\pm 1) \oplus g &\mapsto \{\pm g\} \end{aligned}$$

under the identification

$$\pi_1(BG_+ \wedge \mathbb{K}_{\mathbb{Z}}^{\text{free}}) \cong \{1, -1\} \oplus G_{ab}.$$

See [\[Wal78b, Assertion 15.8 on page 229\]](#). His proof involves rewriting the assembly map in terms of Quillen's Q -construction and then verifying the formula at the simplicial level. Therefore, it seems likely that the following conjecture is true. But a rigorous proof is not known to the author.

Conjecture 5.51 (The Universal Assembly on Fundamental Group)

Let R be a regular ring, and G be a group. The universal assembly map on π_1 is given by the formula

$$K_1(i) \oplus \widehat{\Phi}_1 : K_1(R) \oplus [G_{ab} \otimes K_0(R)] \rightarrow K_1(R[G]), \quad (5.6.7)$$

where $i : R \rightarrow R[G]$ is the inclusion, and the map $\widehat{\Phi}_1$ is induced by the map

$$\begin{aligned} G \times K_0(R) &\rightarrow K_1(R[G]) \\ (g, [P]) &\mapsto \left\{ \widehat{\mathfrak{h}}_g : \begin{array}{ccc} P \otimes_R R[G] & \rightarrow & P \otimes_R R[G] \\ x \otimes u & \mapsto & x \otimes ug \end{array} \right\}. \end{aligned} \quad (5.6.8)$$

Again, we have $\{\widehat{\mathfrak{h}}_g\}$ instead of $\{\mathfrak{h}_g\}$ as in [Corollary 3.9](#) because of [Proposition 2.7](#).

5.7. A Final Remark on Extending the Loday Pairing. One might ask why we use the Gersten-Wagoner delooping to extend the Loday pairing

$$\gamma_{\text{Loday}} : BGL(R)^+ \wedge BGL(S)^+ \rightarrow BGL(R \otimes S)^+$$

to

$\gamma'_{\text{Loday}} : [K_0(R) \times BGL(R)^+] \wedge [K_0(S) \times BGL(S)^+] \rightarrow K_0(R \otimes S) \times BGL(R \otimes S)^+$,
instead of using the “naive approach”:

$$\begin{aligned} \gamma_{\text{naive}} : [K_0(R) \times BGL(R)^+] \wedge [K_0(S) \times BGL(S)^+] &\rightarrow K_0(R \otimes S) \times BGL(R \otimes S)^+ \\ ([P], x) \wedge ([Q], y) &\mapsto ([P \otimes Q], \gamma_{\text{Loday}}(x, y)). \end{aligned} \quad (5.7.1)$$

However, if we look at the induced product map

$$\begin{aligned} \star_{\text{naive}} : K_i(R) \otimes K_j(S) &\rightarrow K_{i+j}(R \otimes S) \\ [f] \otimes [g] &\mapsto [\gamma_{\text{naive}} \circ (f \wedge g)], \end{aligned} \quad (5.7.2)$$

we immediately see that it is the zero map when $i = 0$ and $j > 0$. Therefore, algebraically the naive pairing gives the wrong thing.

Secondly, if the naive pairing extends to a pairing (or weak pairing)

$$\gamma_{\text{naive}} : \mathbb{K}_R^{\text{GW}} \wedge \mathbb{K}_S^{\text{GW}} \rightarrow \mathbb{K}_{R \otimes S}^{\text{GW}} \quad (5.7.3)$$

of spectra, then one can define an assembly map

$$\alpha_{\text{naive}} : BG_+ \wedge \mathbb{K}_R^{\text{GW}} \rightarrow \mathbb{K}_{R[G]}^{\text{GW}} \quad (5.7.4)$$

by mimicking [Definition 3.3](#). However, the cokernel $\text{coker}(\pi_1(\alpha_{\text{naive}}))$ of this assembly on π_1 will not be isomorphic to the classical Whitehead group $\text{Wh}_1(G)$ when $R = \mathbb{Z}$. Since everyone agrees that the cokernel of the universal assembly map on π_1 should be isomorphic to $\text{Wh}_1(G)$, the assembly map α_{naive} is not the correct one. Therefore, topologically the naive pairing gives the wrong thing.

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